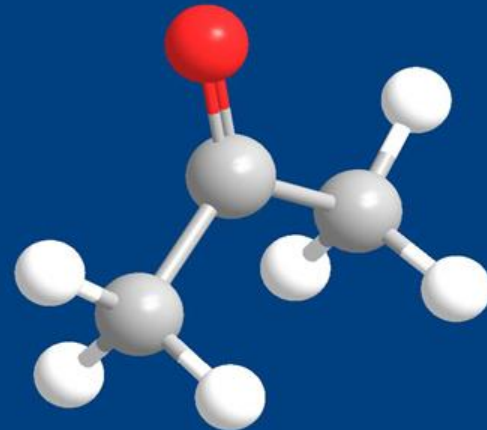
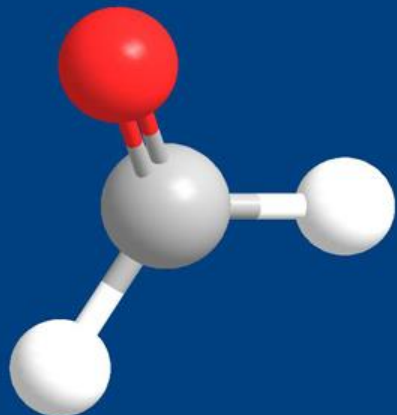
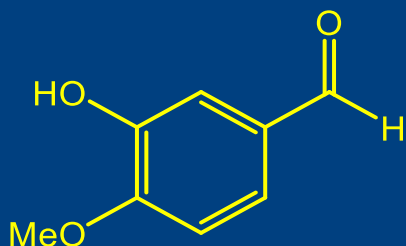




# 第十章 醛和酮

## Aldehydes and Ketones



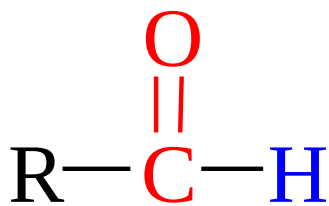
Vanillin  
香草醛 (香兰素)



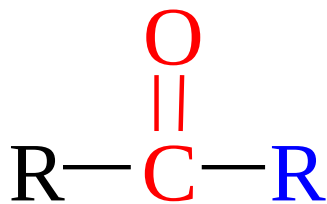
Camphor  
樟脑

# 醛酮结构

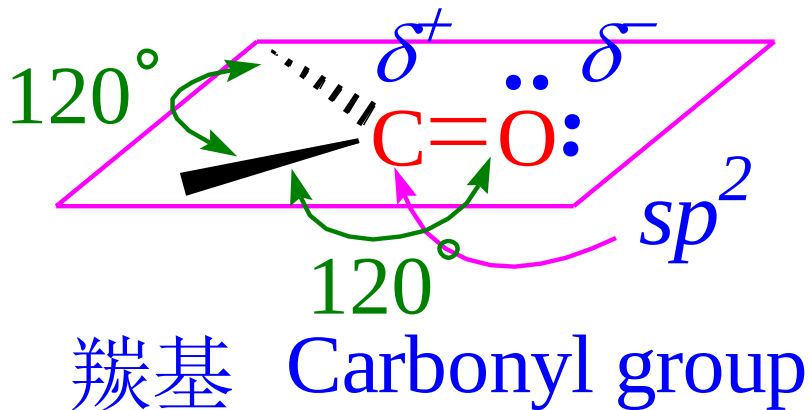
醛、酮的官能团是羰基，羰基碳原子为 $sp^2$ 杂化，带部分正电荷，是极性官能团



醛



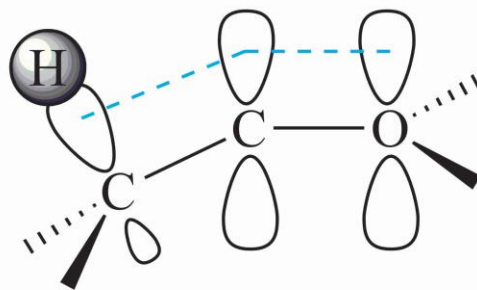
酮



## Acidic $\alpha$ -hydrogens



羰基的吸电子诱导效应



$\alpha$ 碳氢键对羰基的超共轭效应



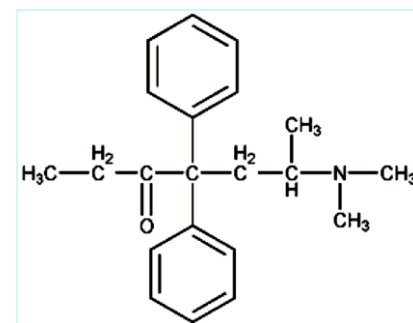
➤ 广泛存在于自然界中, 可从动植物体内提取

柠檬醛、薄荷酮、樟脑、麝香

雄酮激素、雌酮激素、可的松、炔诺酮、视黄醛

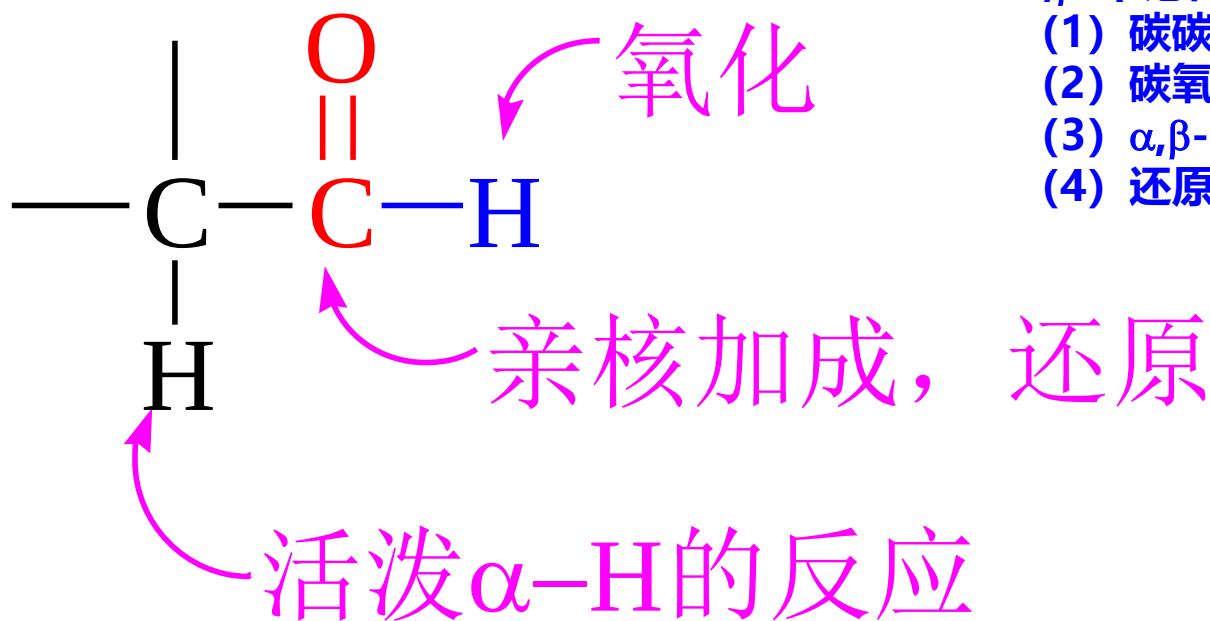
## 应用

- 有机合成的重要原料、溶剂，如丙酮、乙醛等；
- 香料、香精，如月桂醛、柠檬醛、茴香醛等；
- 药物，如美沙酮（戒毒）、盐酸麻黄苯丙酮（冠心病）等。



# 醛酮的亲核加成反应

## 醛酮的反应概貌



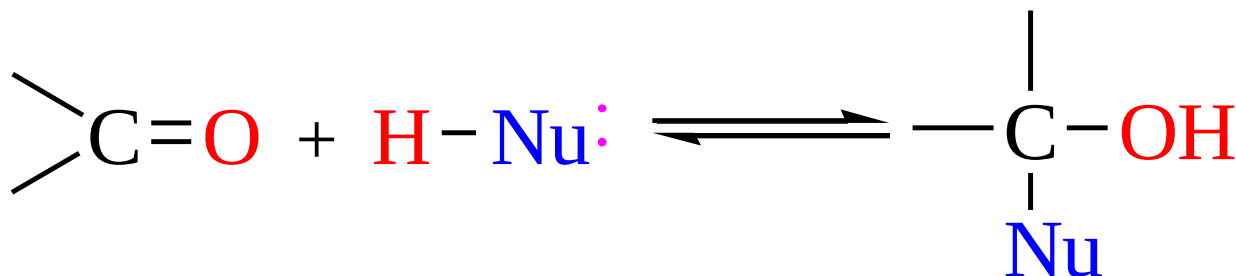
### $\alpha$ -活泼H的反应

- (1) 烯醇化
- (2)  $\alpha$ -卤代 (卤仿反应)
- (3) 醇醛缩合反应

### $\alpha, \beta$ -不饱和醛酮的反应:

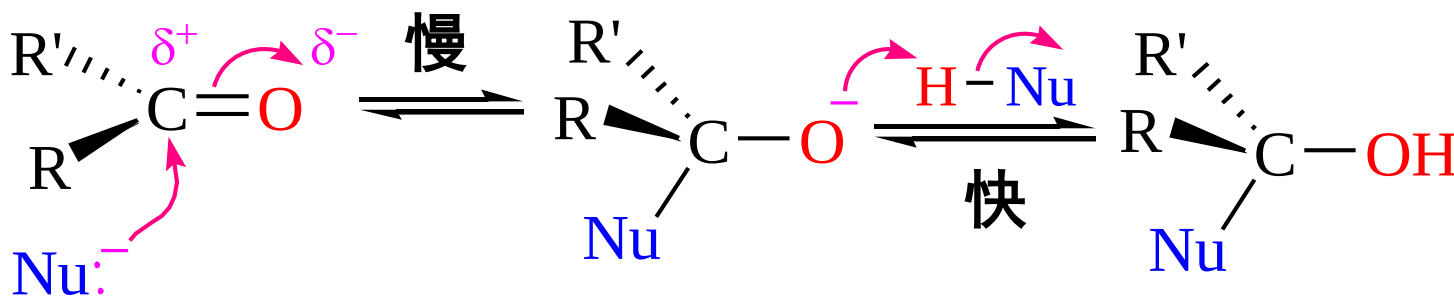
- (1) 碳碳双键的亲电加成
- (2) 碳氧双键的亲核加成
- (3)  $\alpha, \beta$ -不饱和醛酮的共轭加成
- (4) 还原

# 羰基的亲核加成



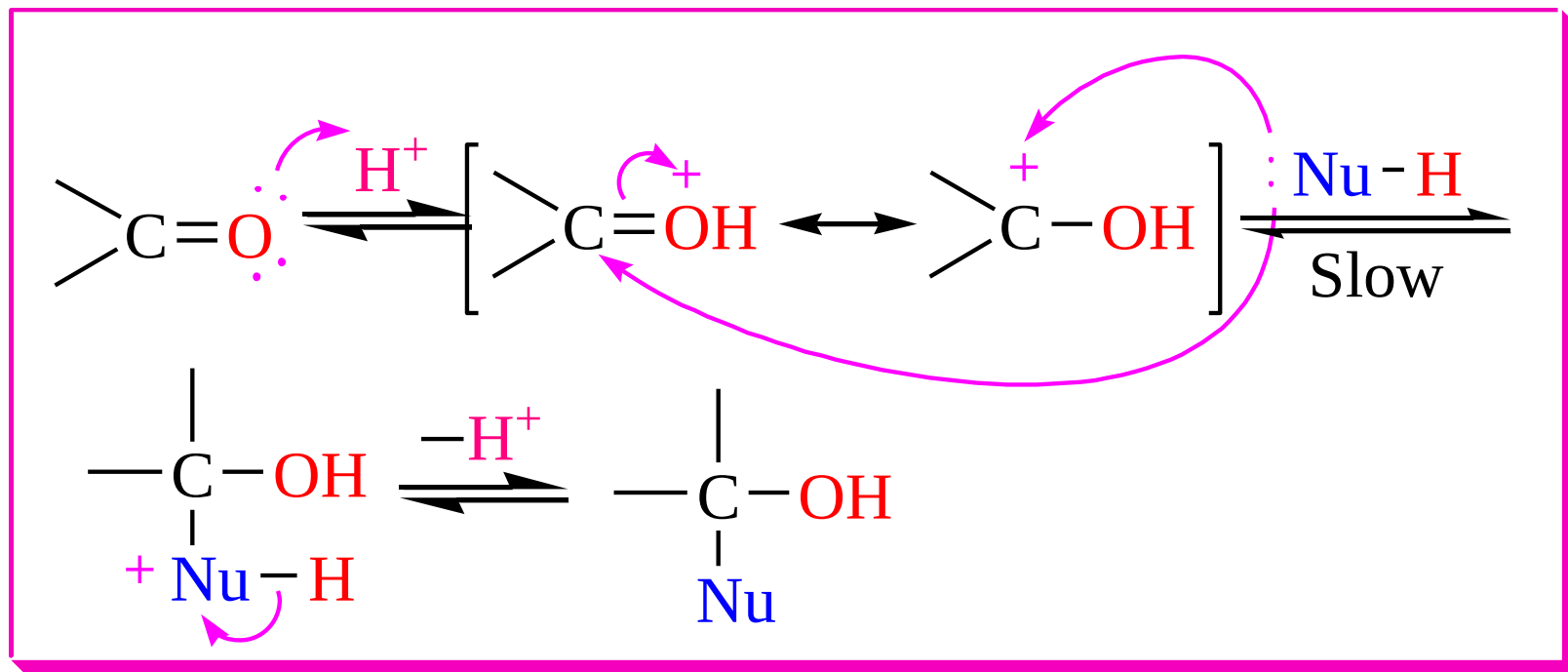
## 反应机理

### 1) 无催化剂或碱性条件下的加成机理



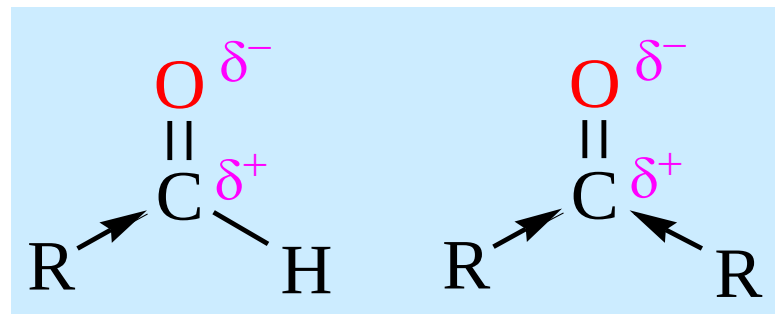
# 羰基的亲核加成

## 2) 酸催化下的反应机理



## 3) 反应活性:

醛 > 脂肪酮 > 芳香酮  
环酮 > 脂肪酮



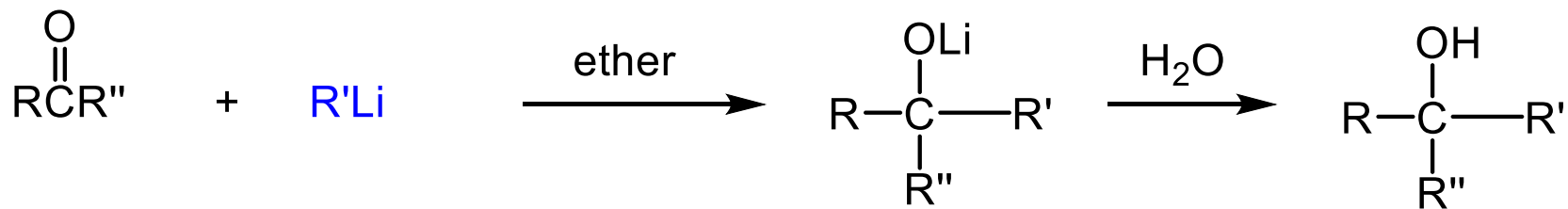
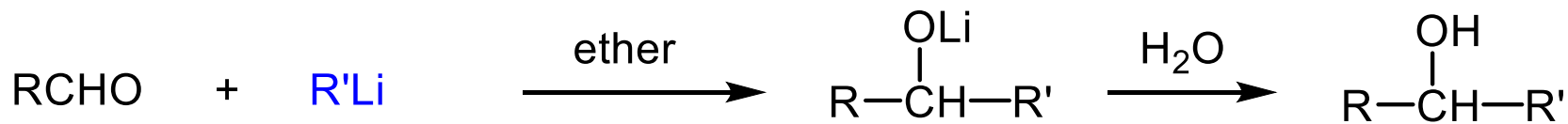
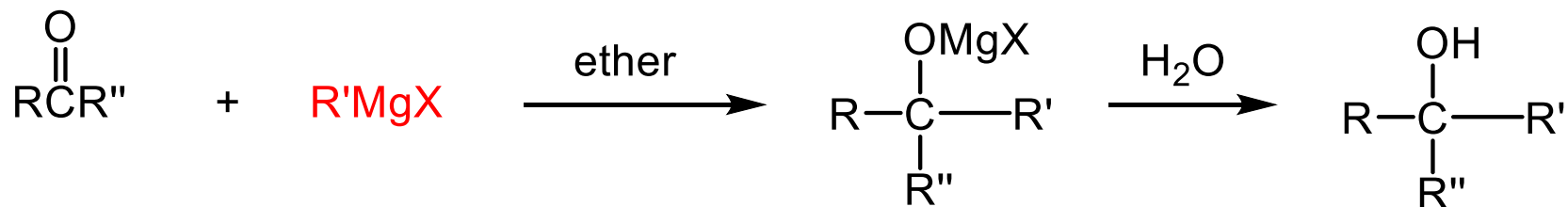
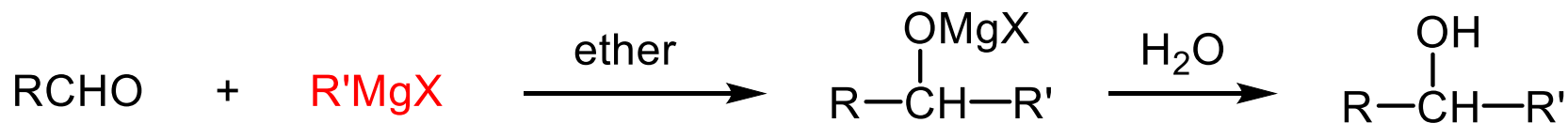
# 羰基与含碳亲核试剂的加成

## 亲核试剂的分类

- 含碳的亲核试剂  $\text{RMgX}$ ,  $\text{HCN}$ ,  $\text{C}^-$
- 含氧的亲核试剂  $\text{H}_2\text{O}$ ,  $\text{ROH}$
- 含硫的亲核试剂  $\text{NaHSO}_3$ ,  $\text{RSH}$
- 含氮的亲核试剂  $\text{RNH}_2$ ,  $\text{R}_2\text{NH}$ ,  $\text{H}_2\text{N-G}$

# 羰基与含碳亲核试剂的加成

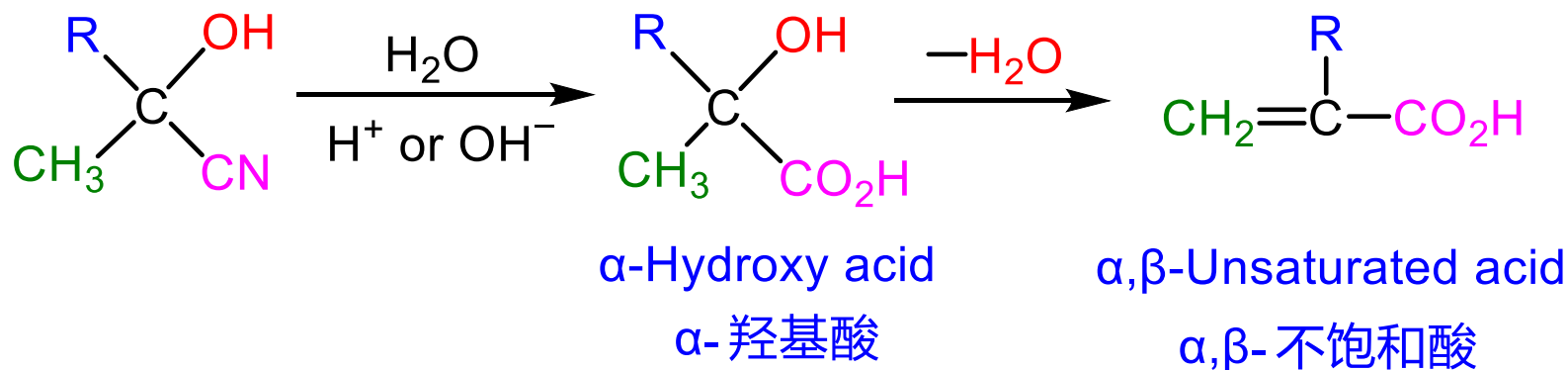
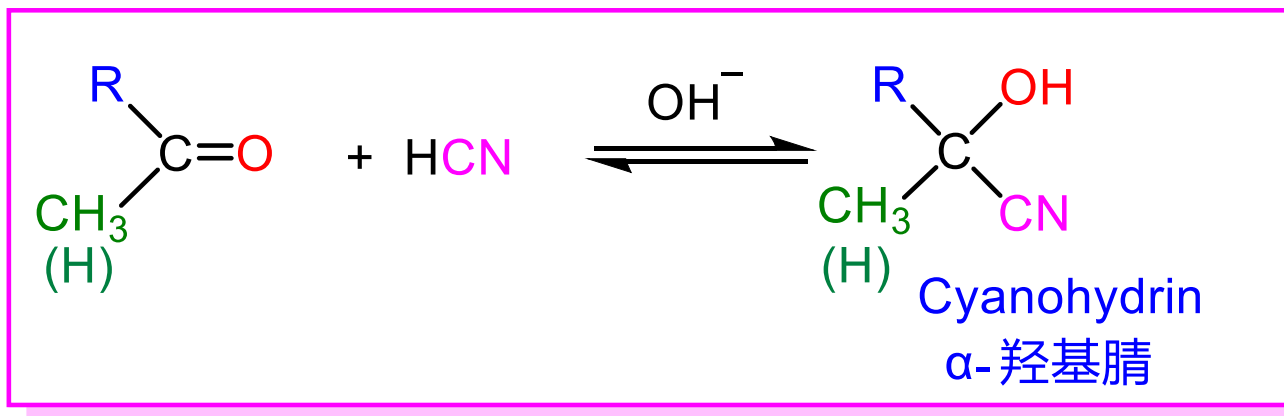
## 1. The Addition of Organometallic Reagents ( $\text{RMgX}$ , $\text{RLi}$ )



制备一级醇：多聚甲醛 (paraformaldehyde)



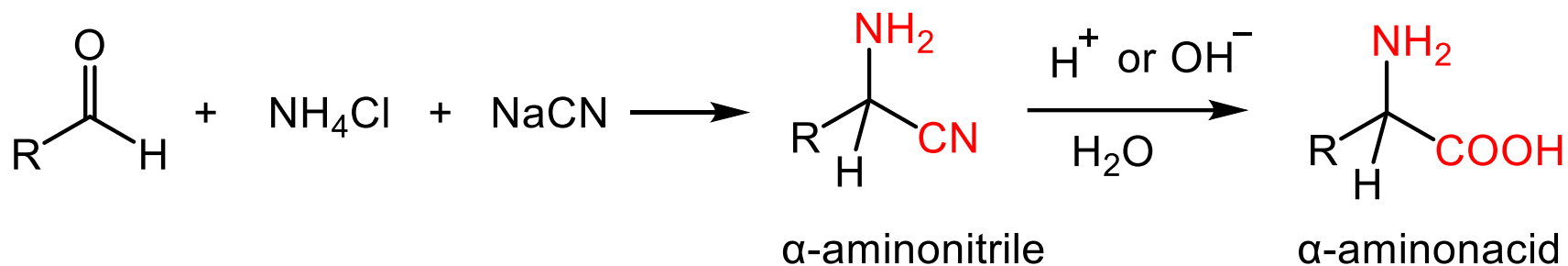
## 2. The Addition of Hydrogen Cyanide (HCN)



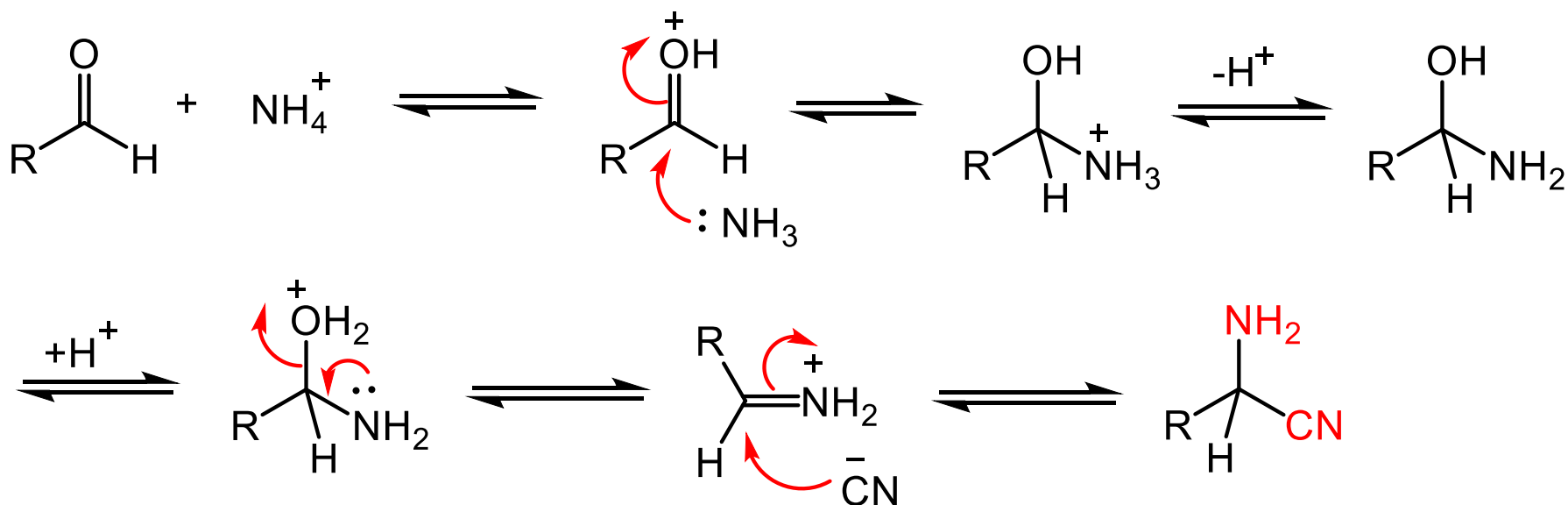
## 斯特雷克氨基酸合成反应

## Strecker反应

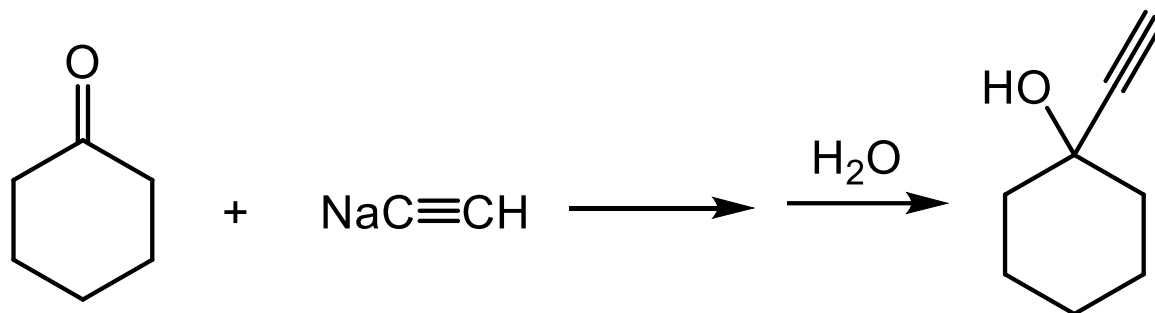
羰基化合物与氯化铵、氰化钠反应，形成 $\alpha$ -氨基腈，经水解可以制备 $\alpha$ -氨基酸：



### Mechanism:

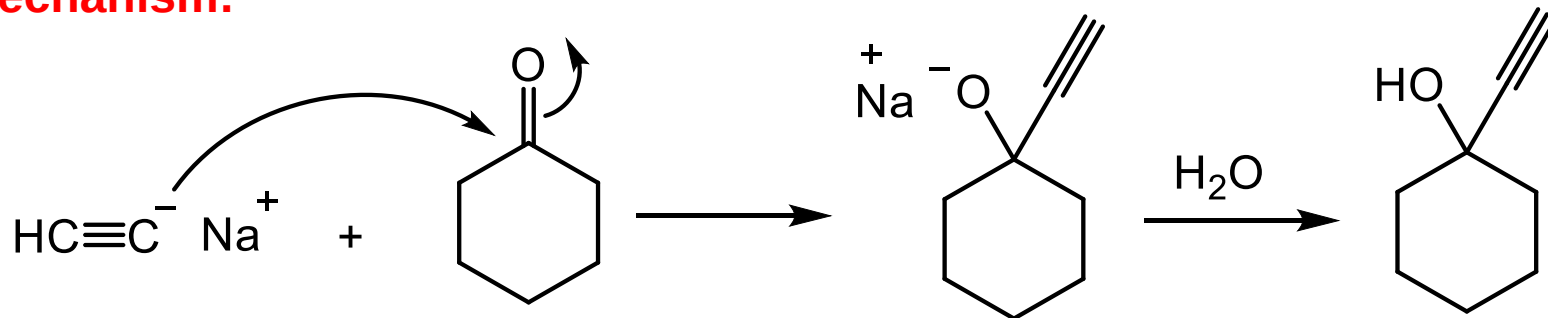


### 3. The Addition of Acetylide (炔化物)

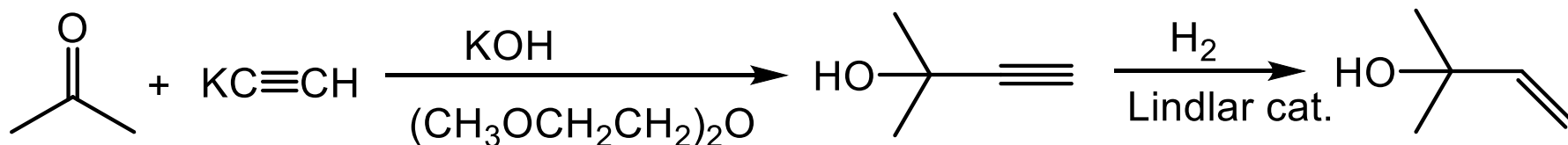


炔化物是很强的亲核试剂。

**Mechanism:**



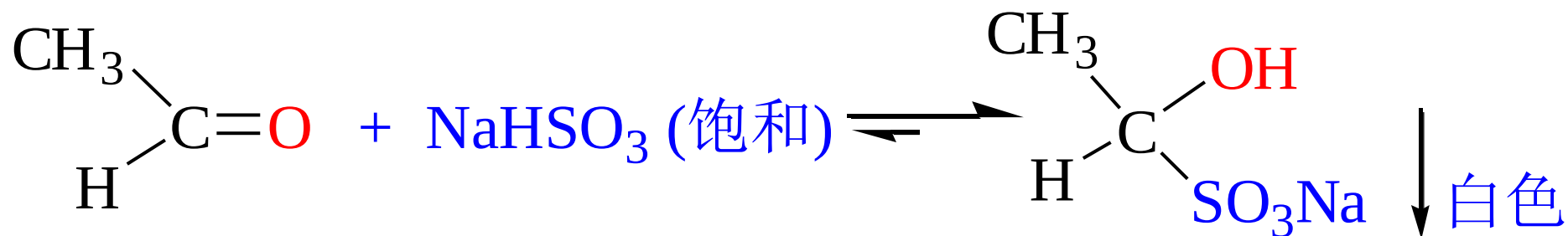
**Example:**



异戊二烯的工业合成路线

# 羰基与含硫亲核试剂的加成

## 与亚硫酸氢钠的反应

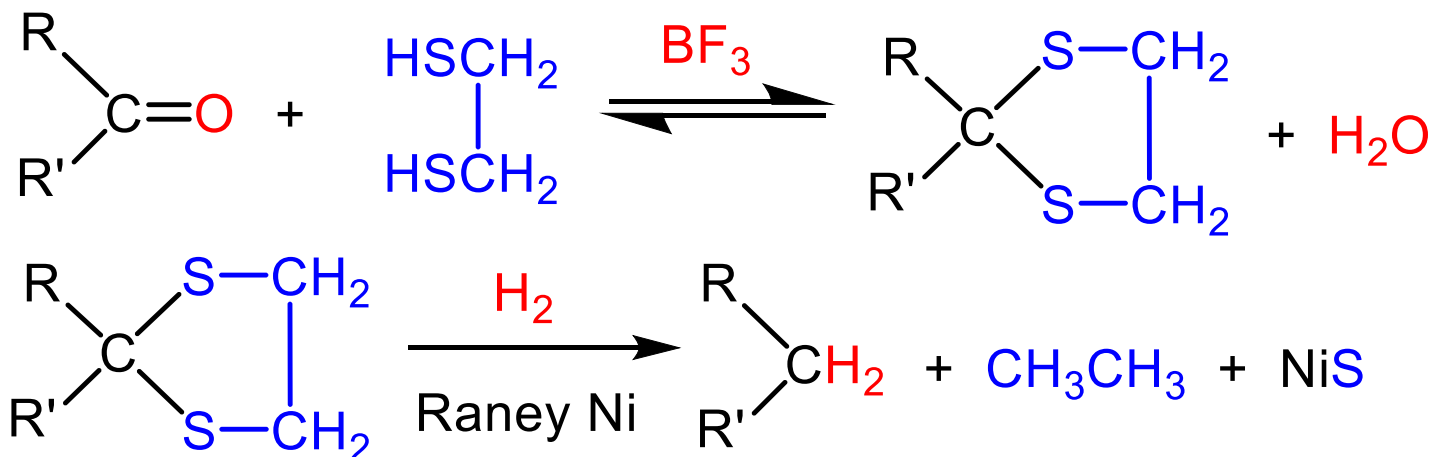


乙醛亚硫酸氢钠加成物

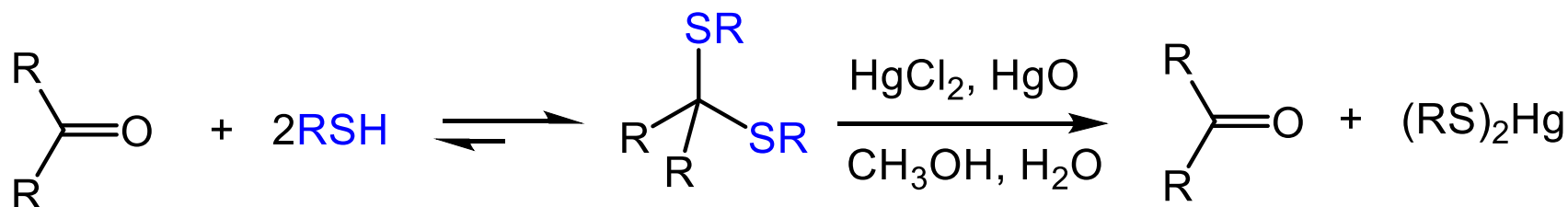
- 醛、脂肪族甲基酮、C8以下环酮能发生反应，其他酮不反应
- 本反应可用于鉴别和分离提纯

## 2. The Addition of Thiol (**RSH**)

强亲核性



脱除醛酮的羰基氧 (还原)

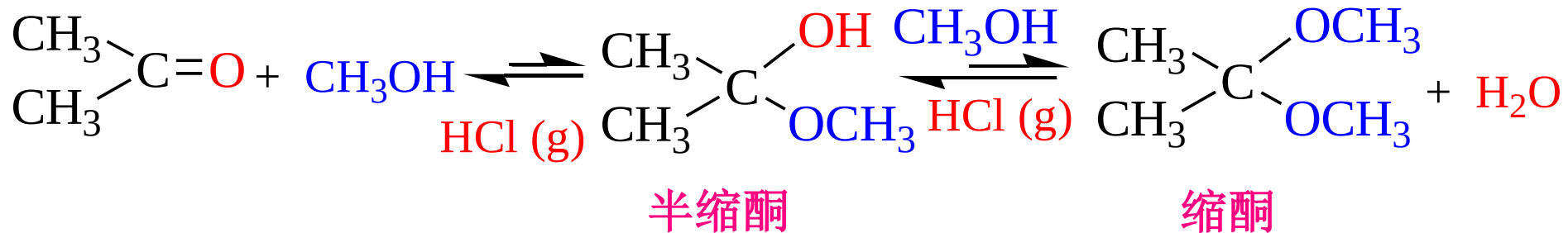
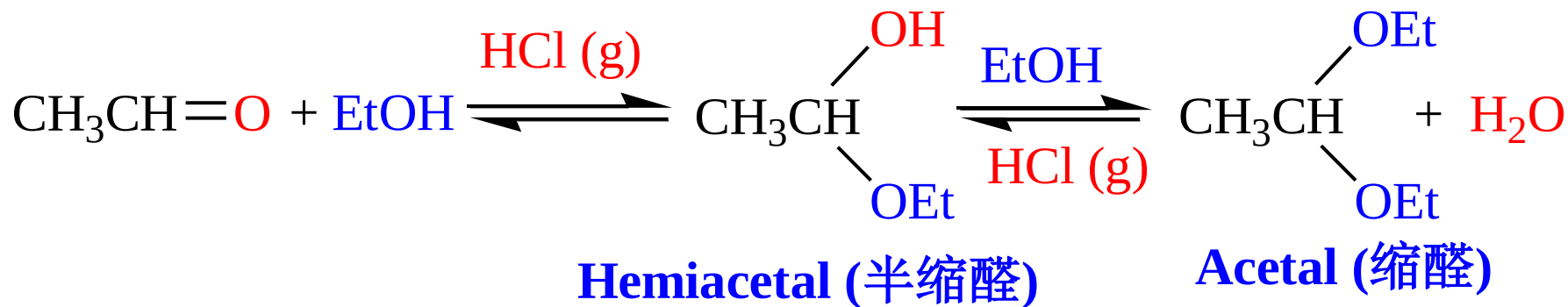


对酸有一定稳定性

保护醛酮的羰基

# 羰基与含氧亲核试剂的加成

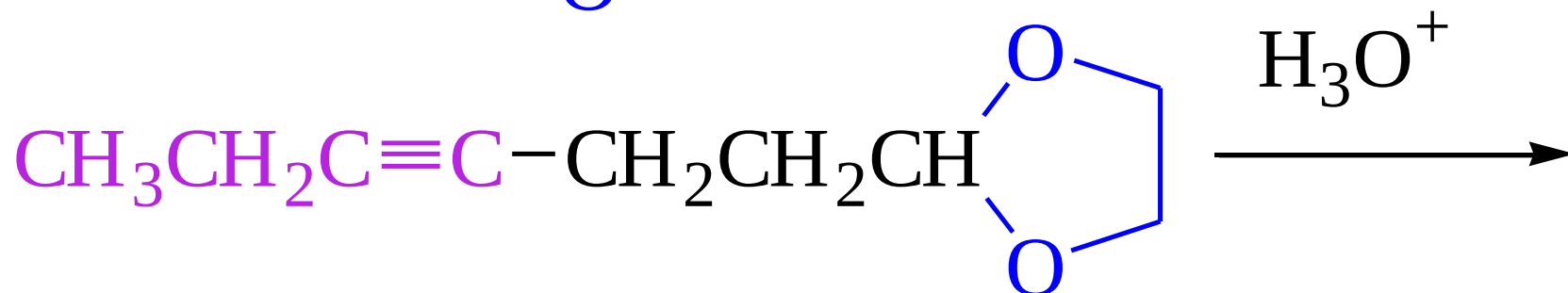
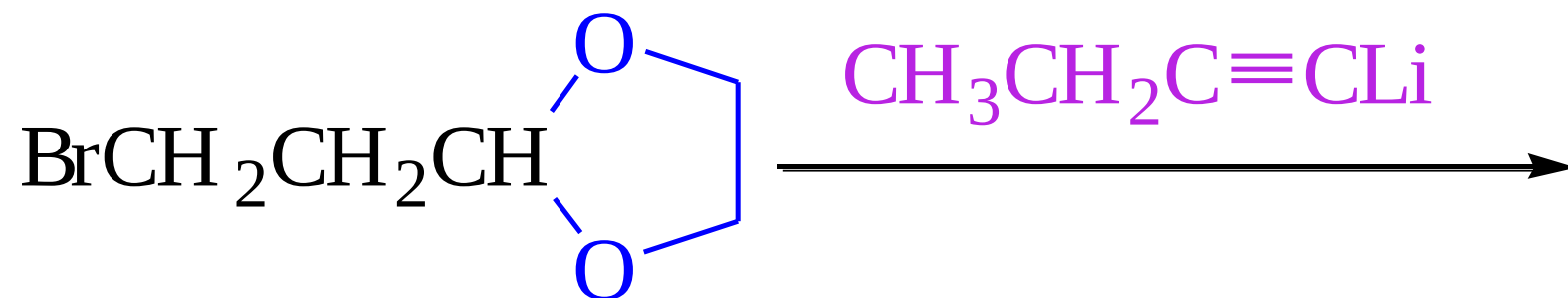
与醇的加成—— 形成半缩醛(酮)，缩醛(酮)



# 反应特点

- 半缩醛(酮)无论在酸性或碱性介质中均不稳定
- 缩醛(酮)在碱性介质中稳定，在酸性介质中不稳定
- 缩醛较易形成，缩酮较难
- 用乙二醇形成环状缩酮反应可提高缩酮的产率

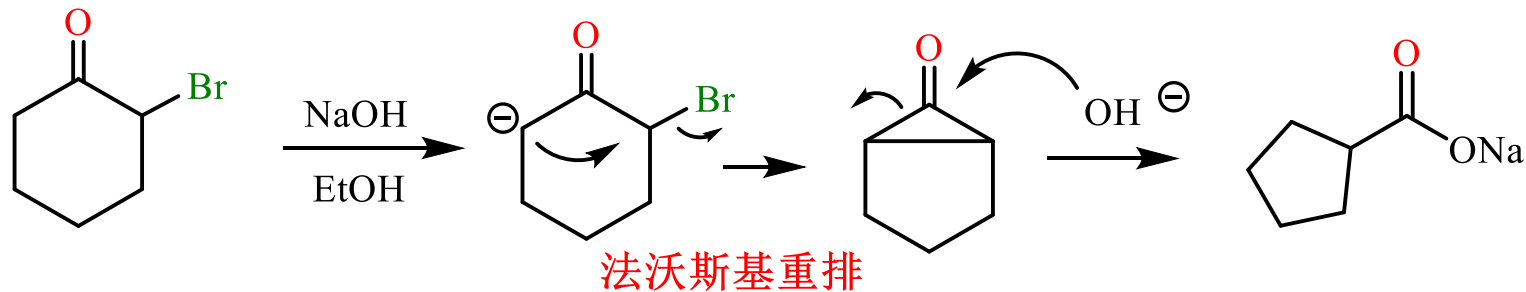
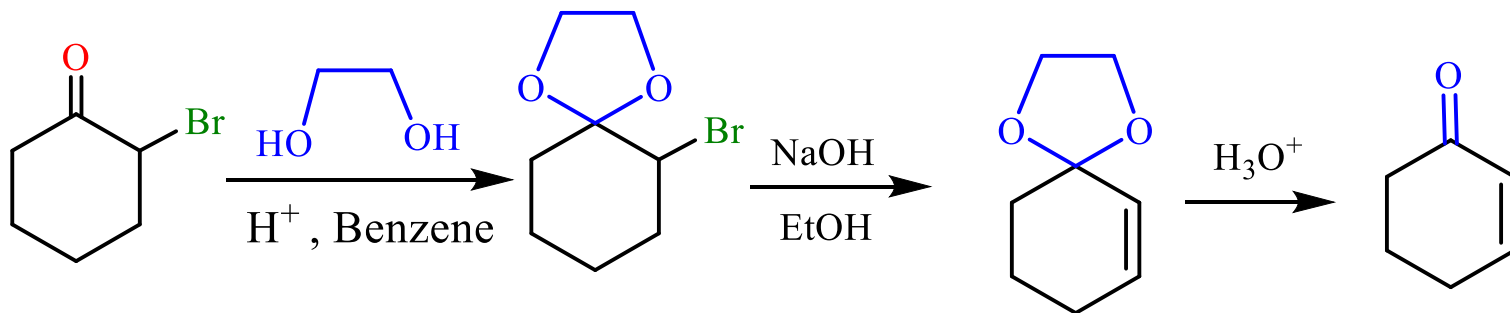
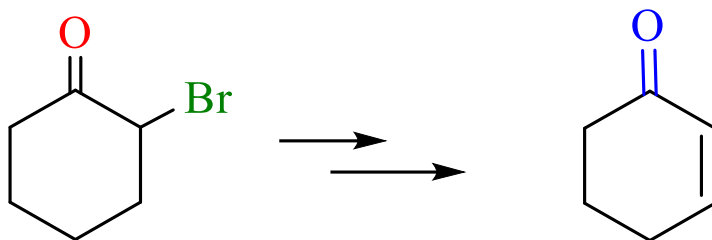
# 应用：保护羰基





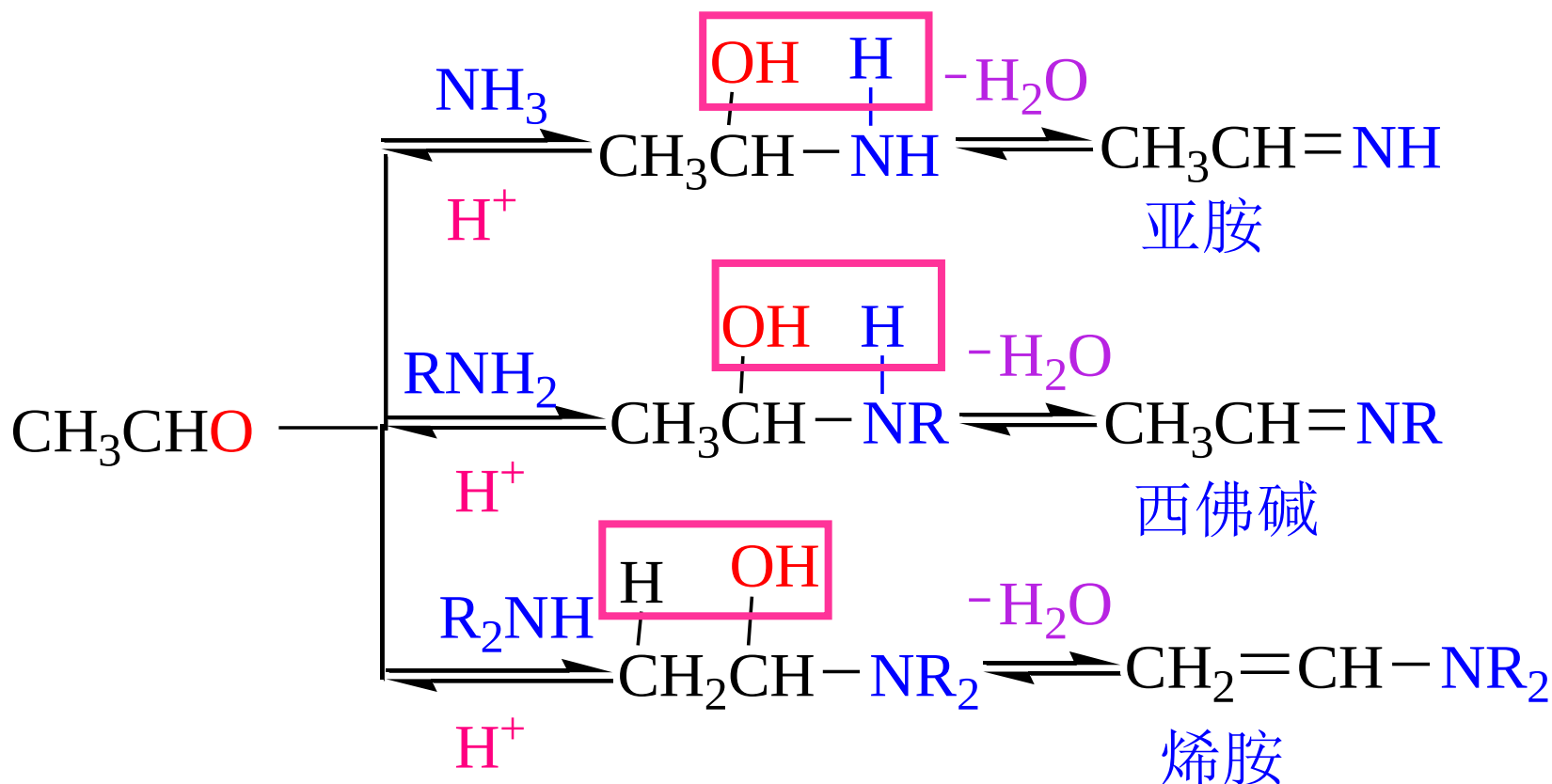
# 测试

请同学们完成下列转换：

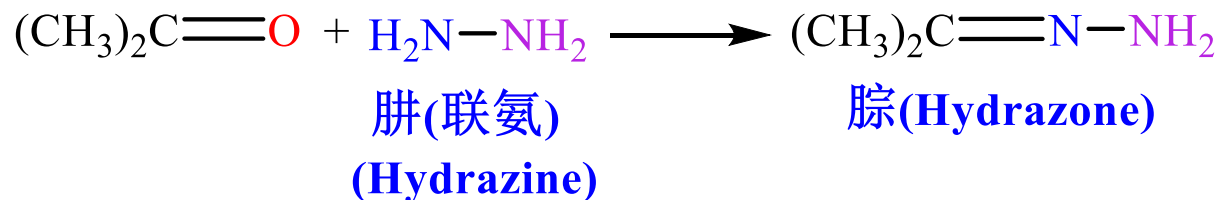
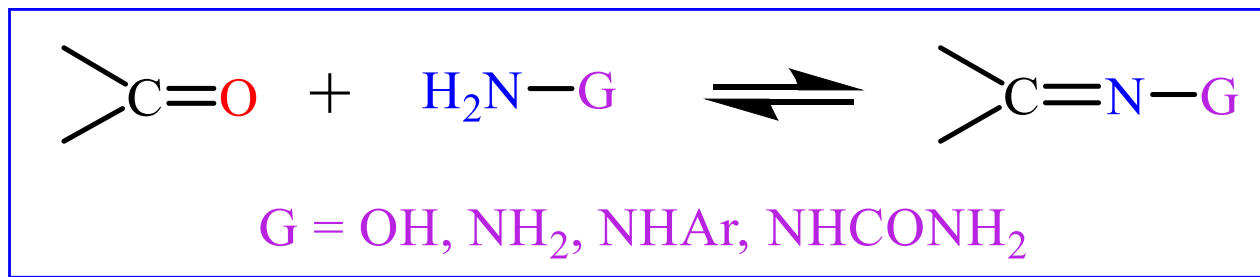


# 羰基与含氮亲核试剂的加成

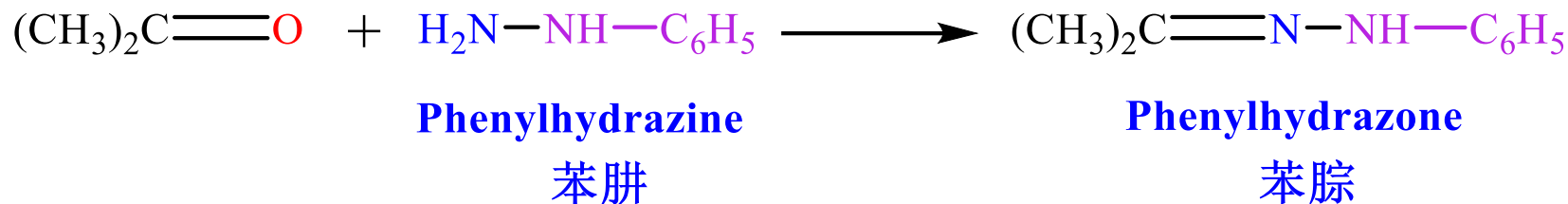
## 1. 与氨或胺的反应 可逆反应，分水操作



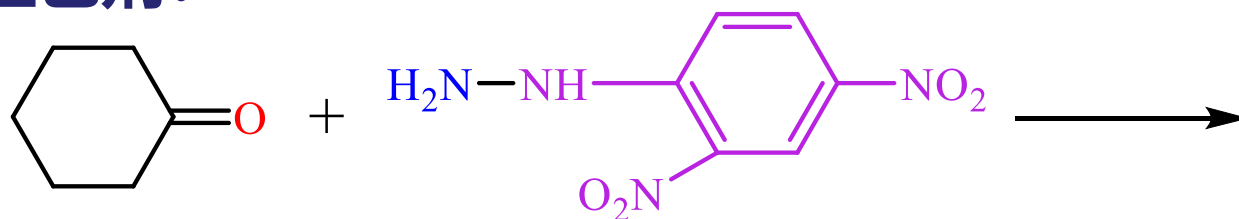
脂肪族Schiff 碱不稳定,芳香族Schiff 碱稳定



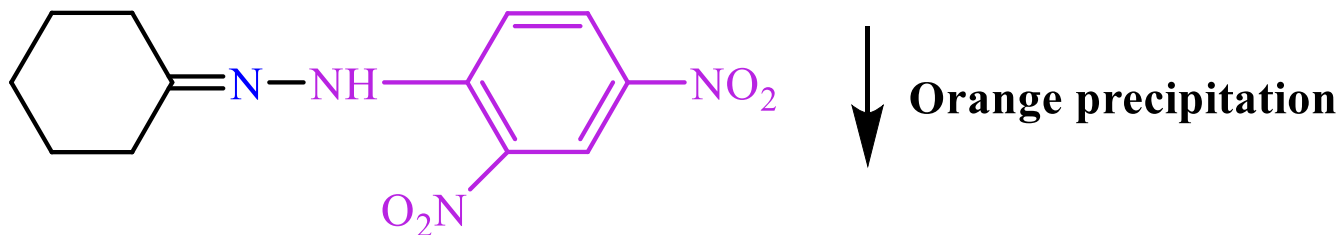
**醛和非对称酮的产物构型如何？ *E*型产物为主**



**醛酮显色剂:**

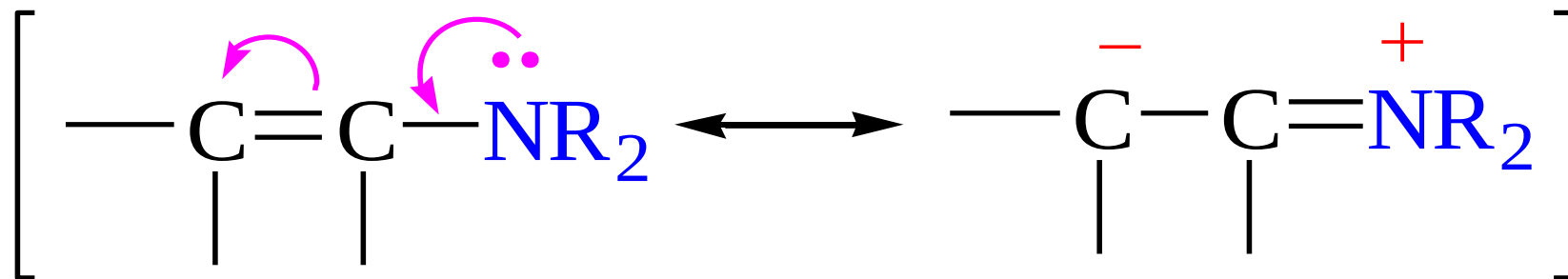


**2,4-Dinitrophenylhydrazine**  
2,4-二硝基苯肼

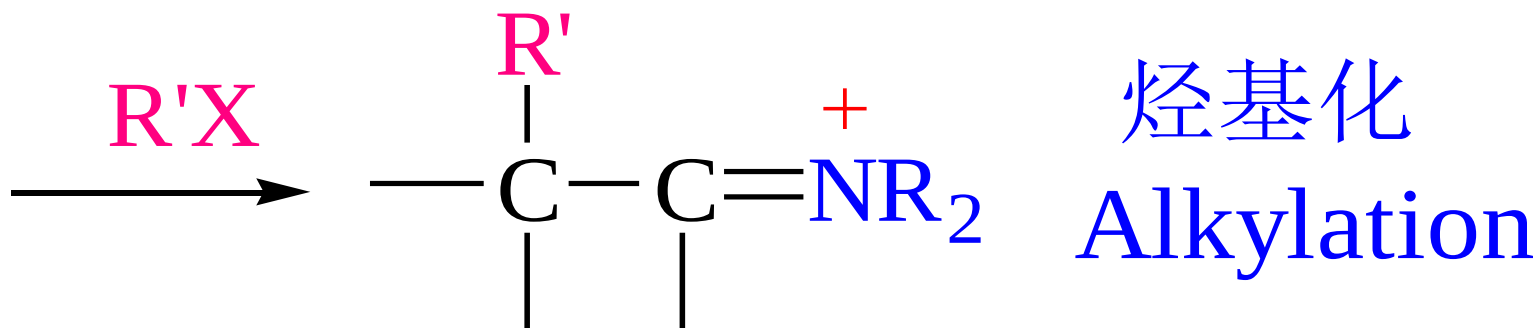


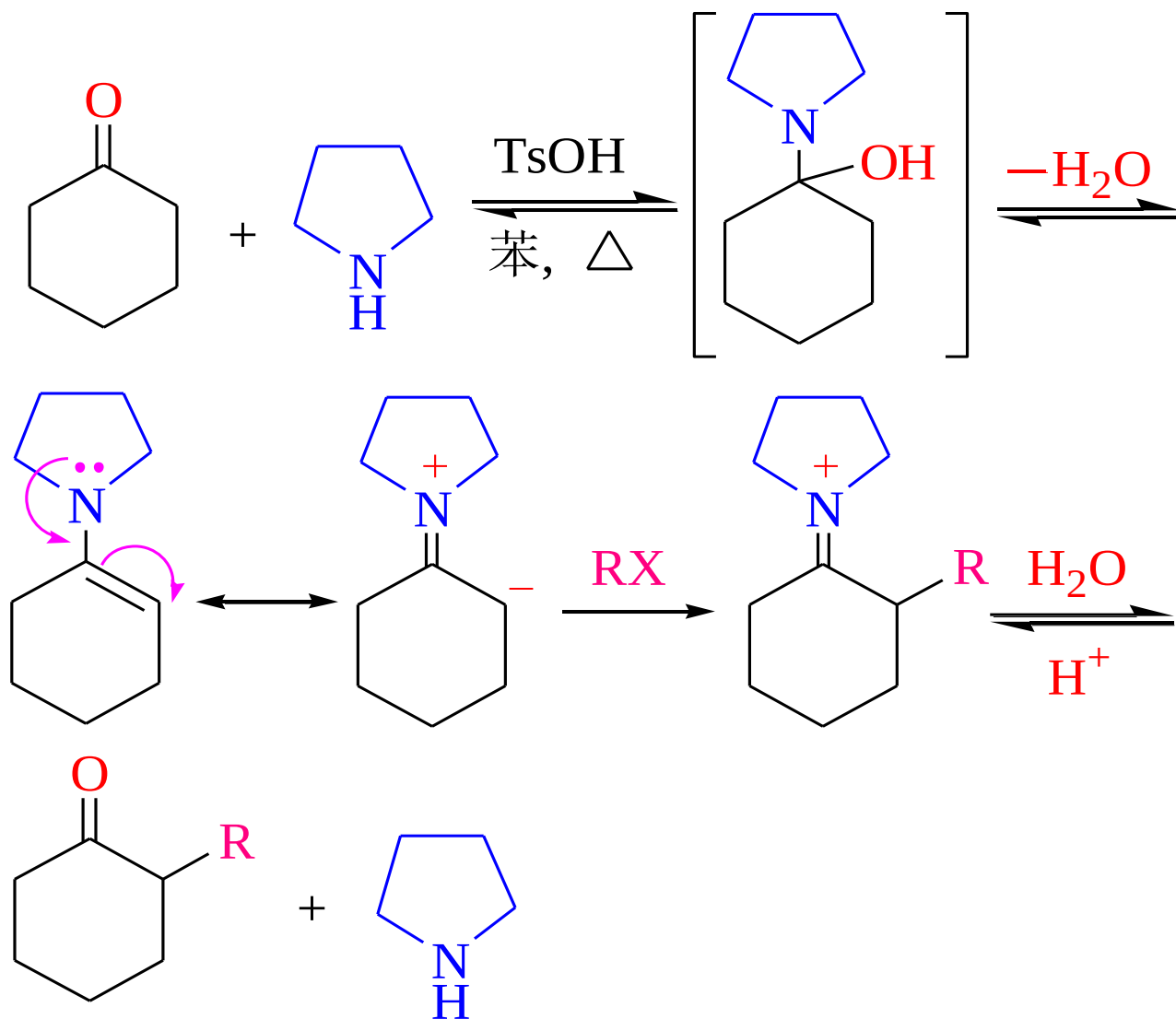
**2,4-Dinitrophenylhydrazone**  
2,4-二硝基苯腙

# 烯胺的性质及应用



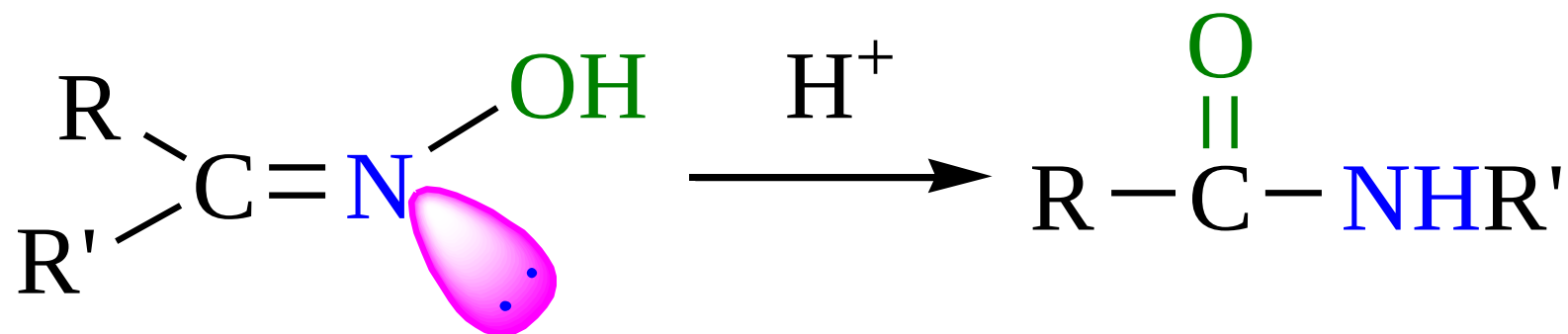
烯胺





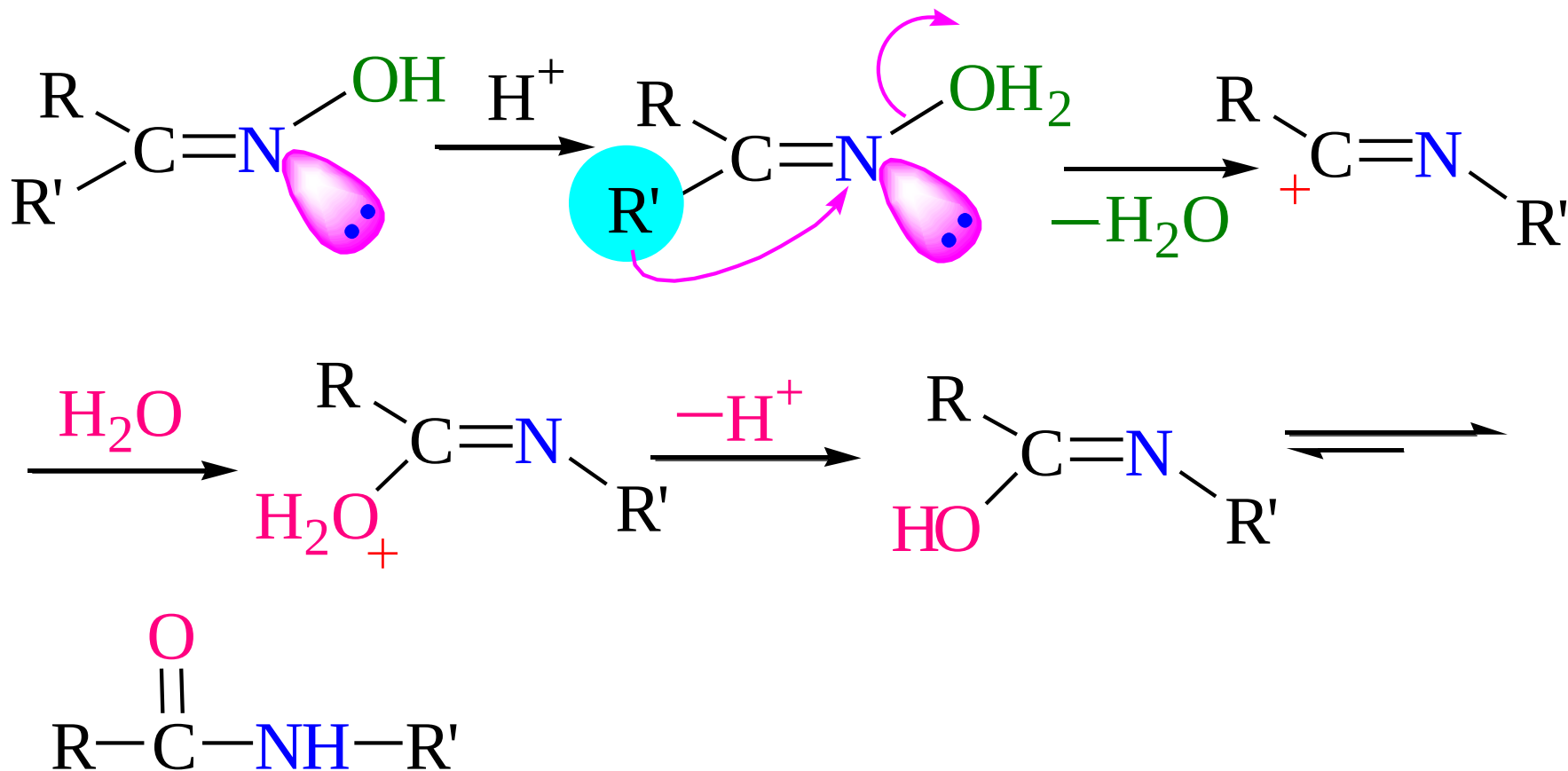
# Beckmann 重排

肟在酸作用下重排形成酰胺



常用催化剂:  $\text{H}_2\text{SO}_4$ ,  $\text{HCl (g)}$ ,

# 反应机理（掌握）

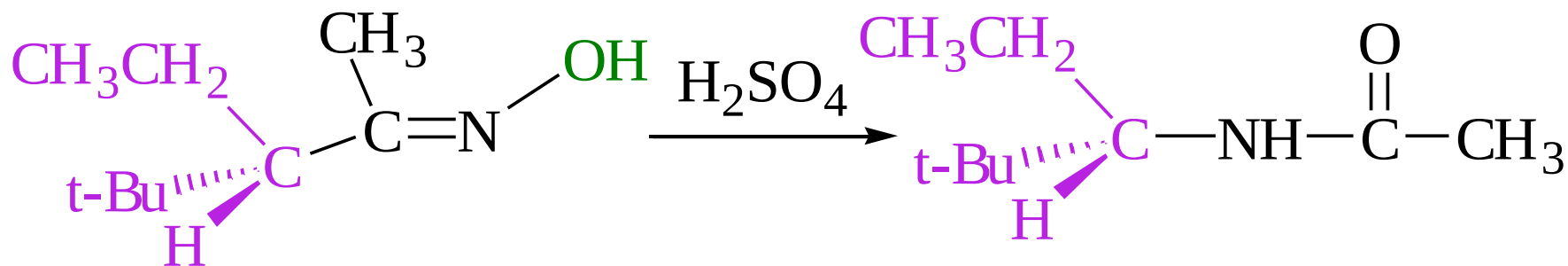




# Beckmann 重排

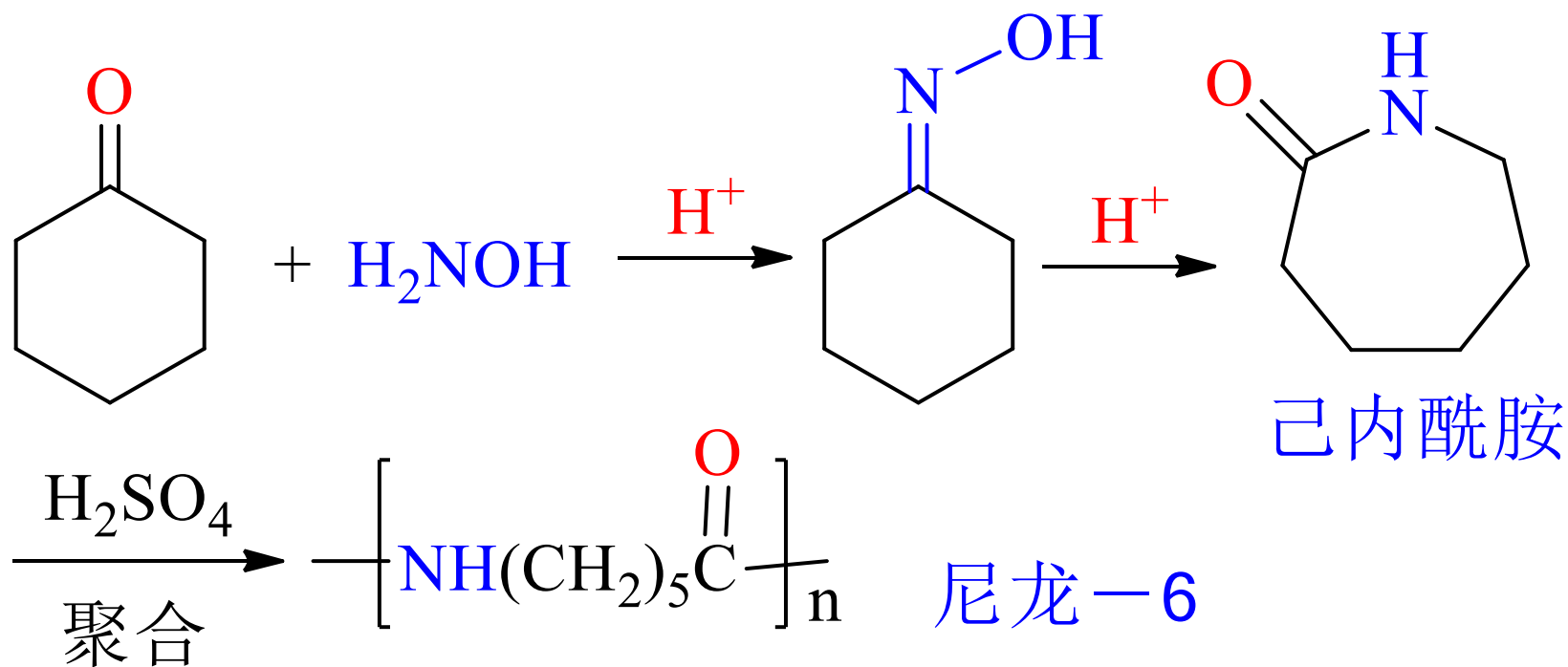
## Beckmann重排的立体化学

- 迁移基团处于羟基的反式位置
- 迁移基团如果是手性碳原子，在迁移过程中其构型保持不变



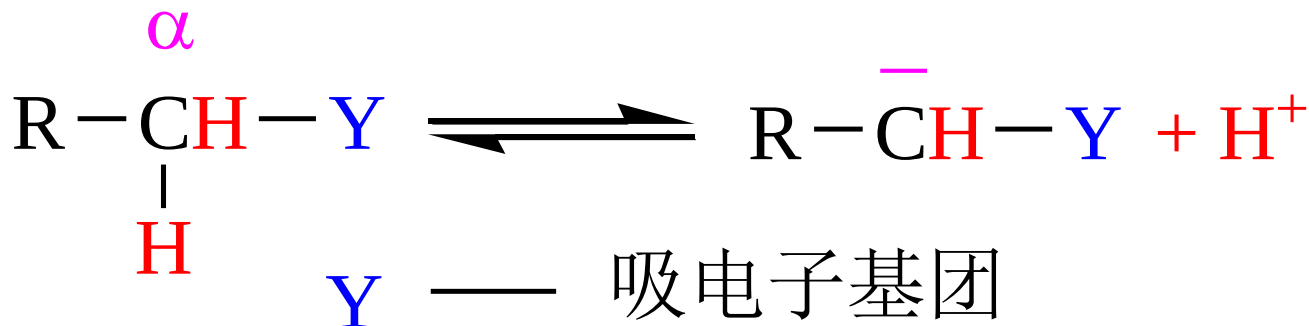
t-Bu, 叔丁基

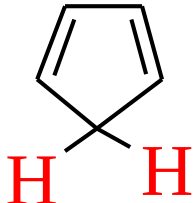
# 应用：由酮制备酰胺



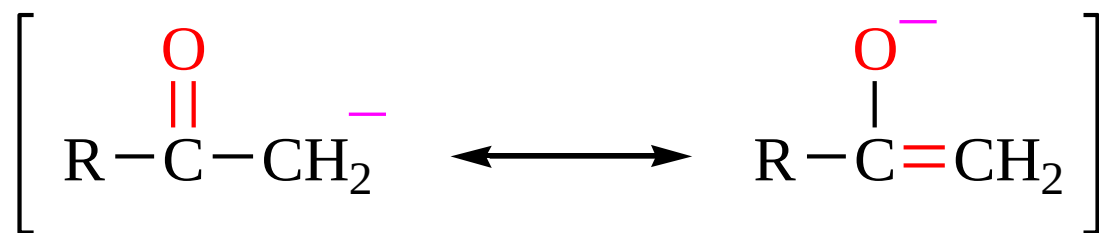
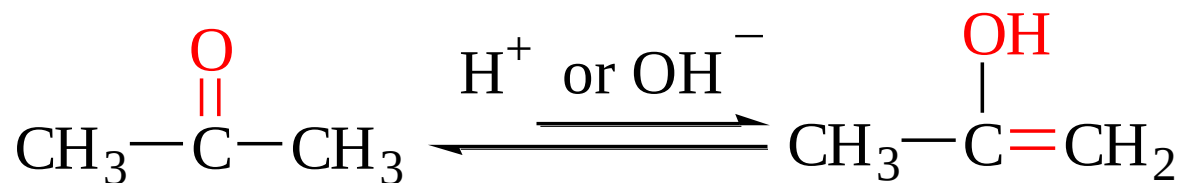
# $\alpha$ -活泼氢的反应

## 1. $\alpha$ -氢的活性



|                                     |                                     |                                                                                       |                          |
|-------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------|--------------------------|
| $\text{CH}_3\text{CH}_3$            | $\text{CH}_2=\text{CHCH}_3$         | $(\text{C}_6\text{H}_5)_2\text{CH}_2$                                                 | $\text{CH}_3\text{CN}$   |
| $\text{pK}_a \sim 50$               | 40                                  | 32.3                                                                                  | 31.3                     |
| $\text{CH}_3\text{SO}_2\text{CH}_3$ | $\text{C}_6\text{H}_5\text{COCH}_3$ |  | $\text{CH}_3\text{NO}_2$ |
| 29                                  | 24.7                                | 16.6                                                                                  | 17.21                    |

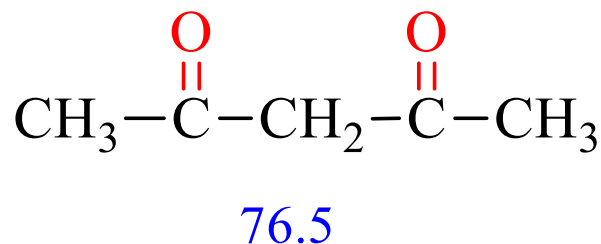
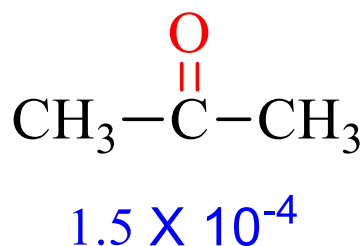
# 酮式和烯醇式——互变异构



碳负离子

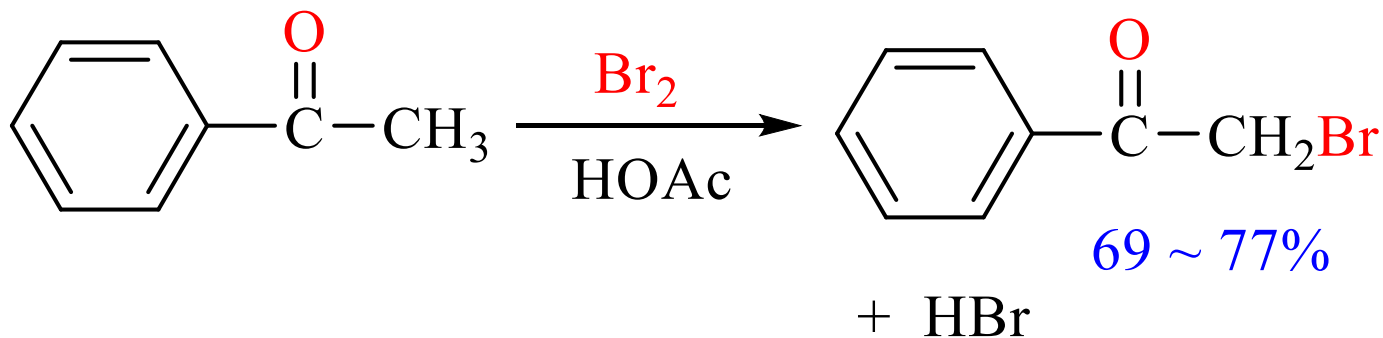
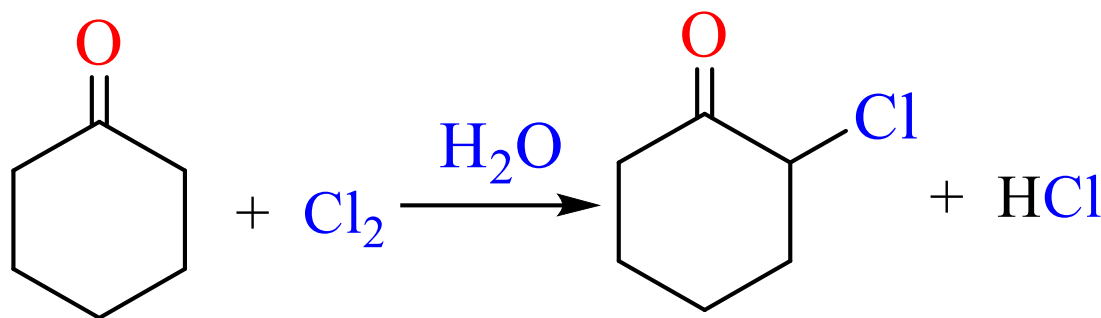
烯醇负离子

烯醇式  
含量

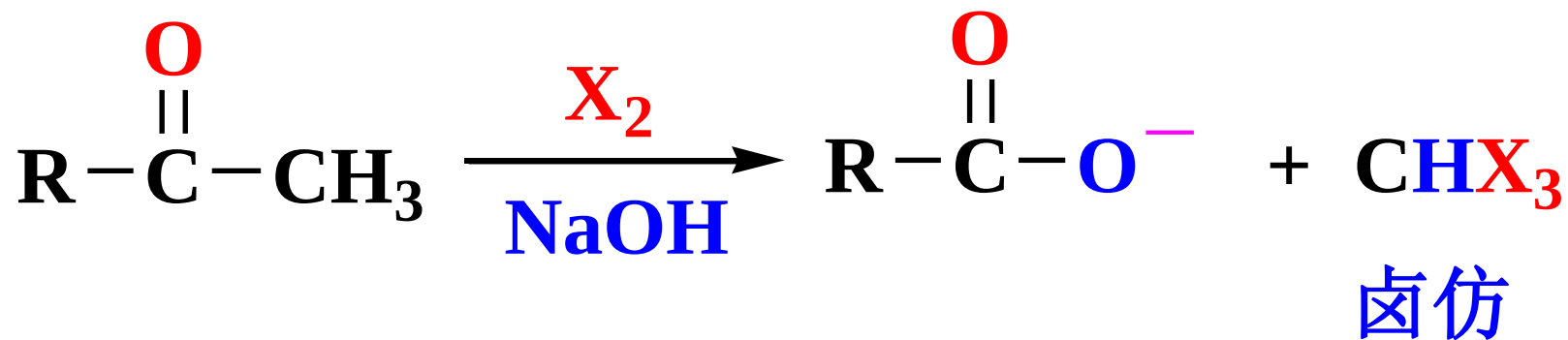


# $\alpha$ -活泼氢的反应

## 2. 卤化和卤仿反应



# 卤仿反应

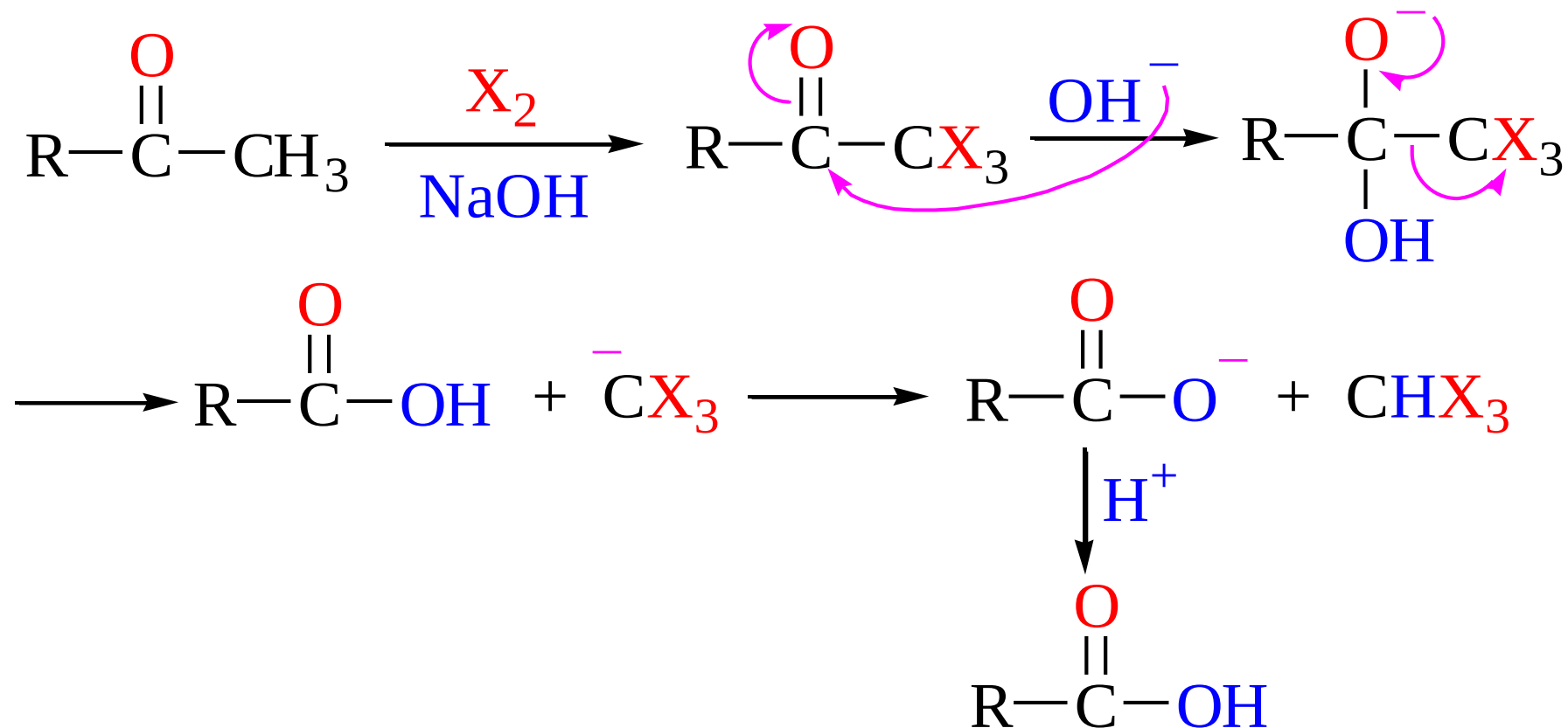


具有  $\text{—}\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_3$   $\text{—}\overset{\text{OH}}{\underset{|}{\text{CH}}}-\text{CH}_3$  结构的

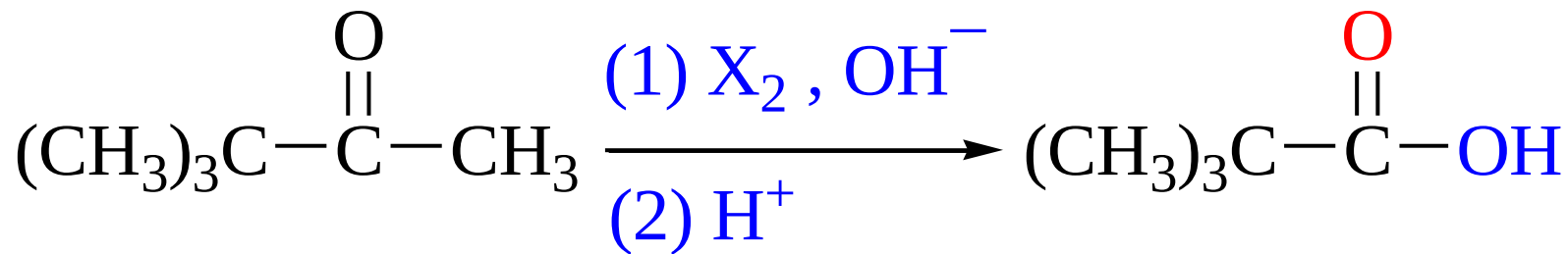
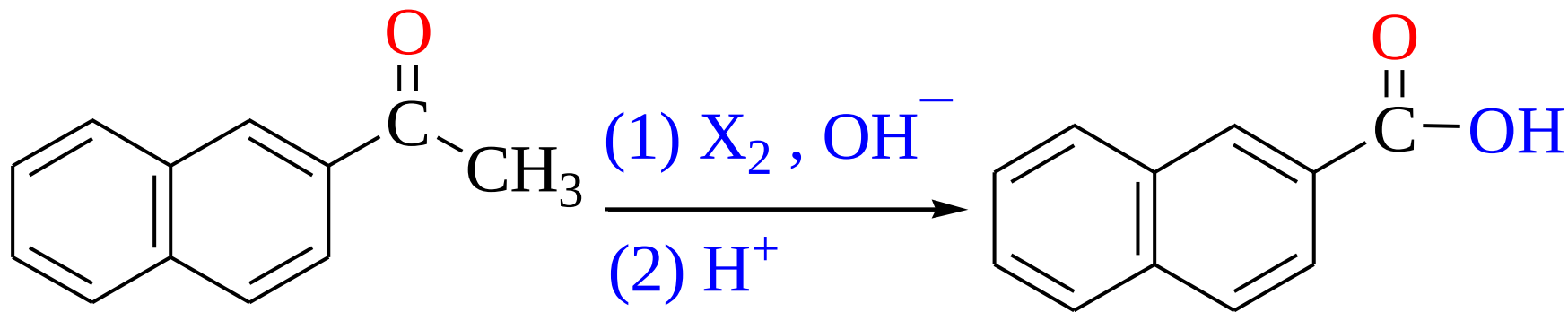
化合物都能发生反应。碘仿反应可用于鉴别。

碘仿 $\text{CHI}_3$  ↓ 黄色

# 反应机理（了解）



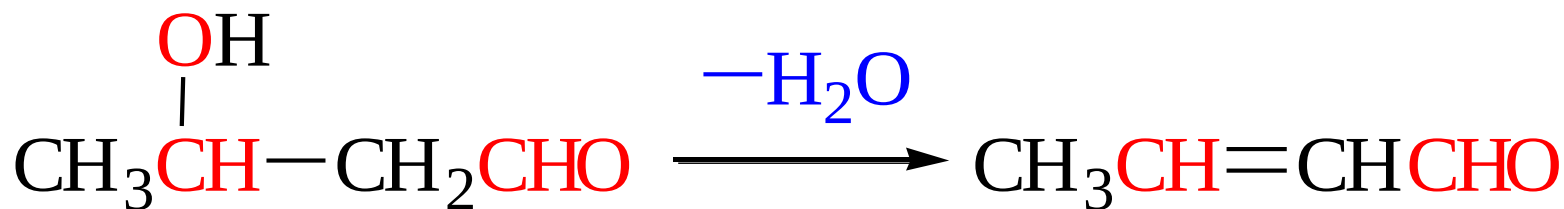
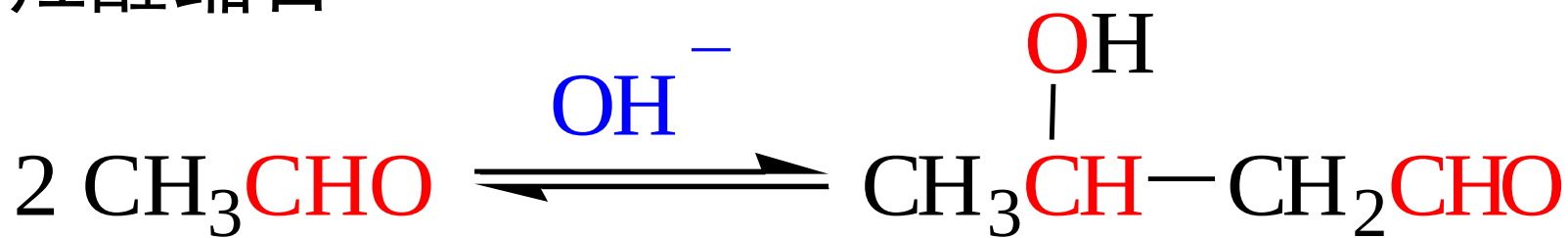
# Example





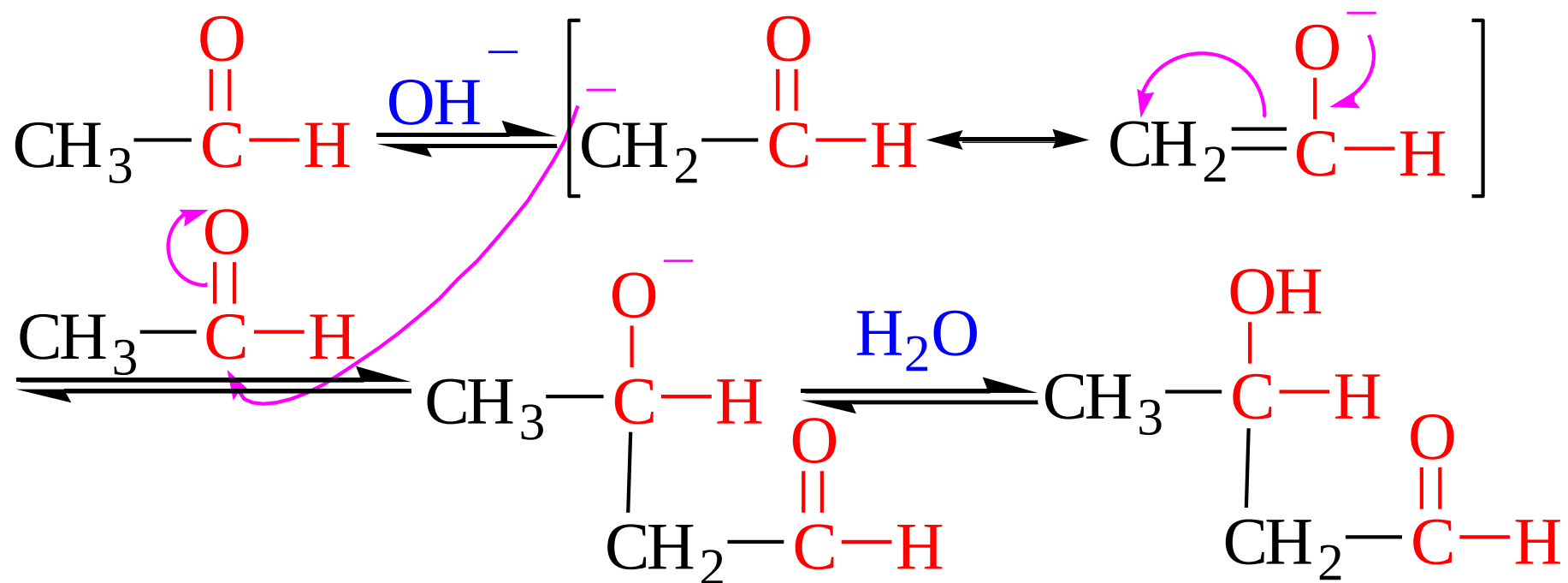
# $\alpha$ -活泼氢的反应

## 3. 羟醛缩合

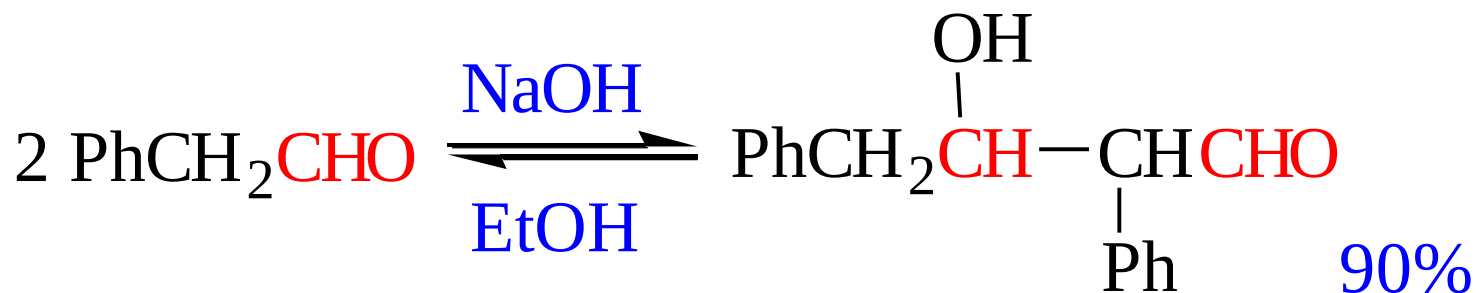
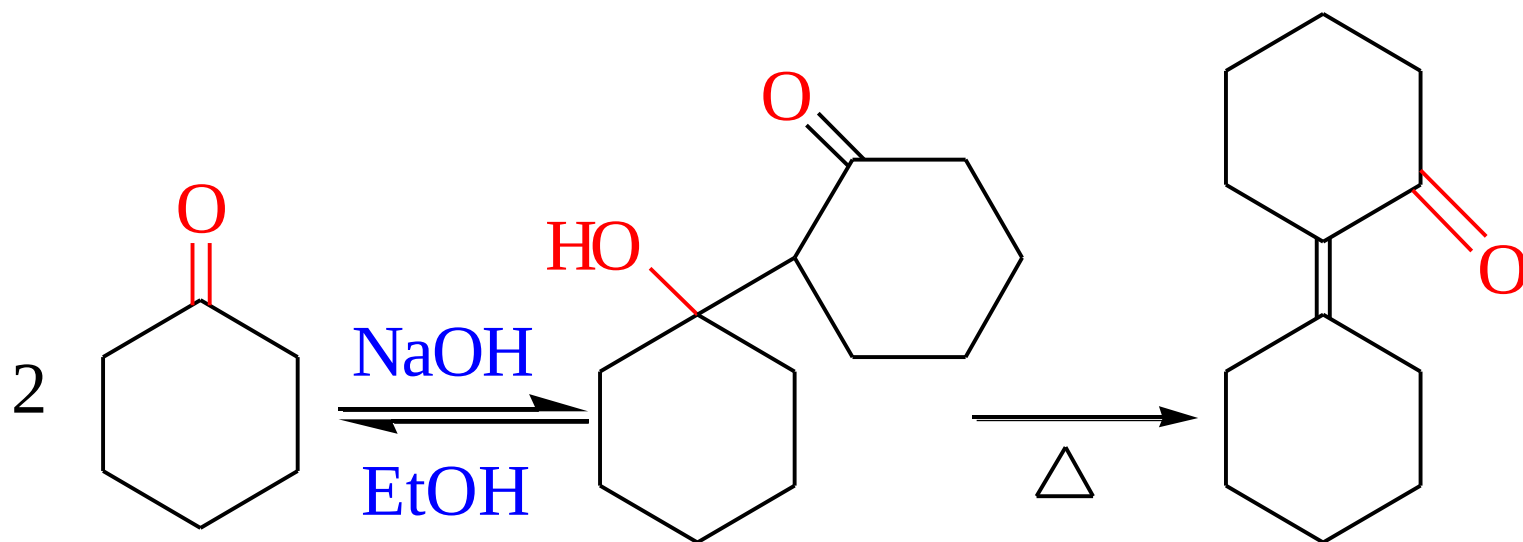


- 含有 $\alpha$ -H的醛(酮)在稀碱的作用下,缩合生成 $\beta$ -羟基醛(酮)
- $\beta$ -羟基醛(酮)易脱水形成 $\alpha, \beta$ -不饱和醛(酮)

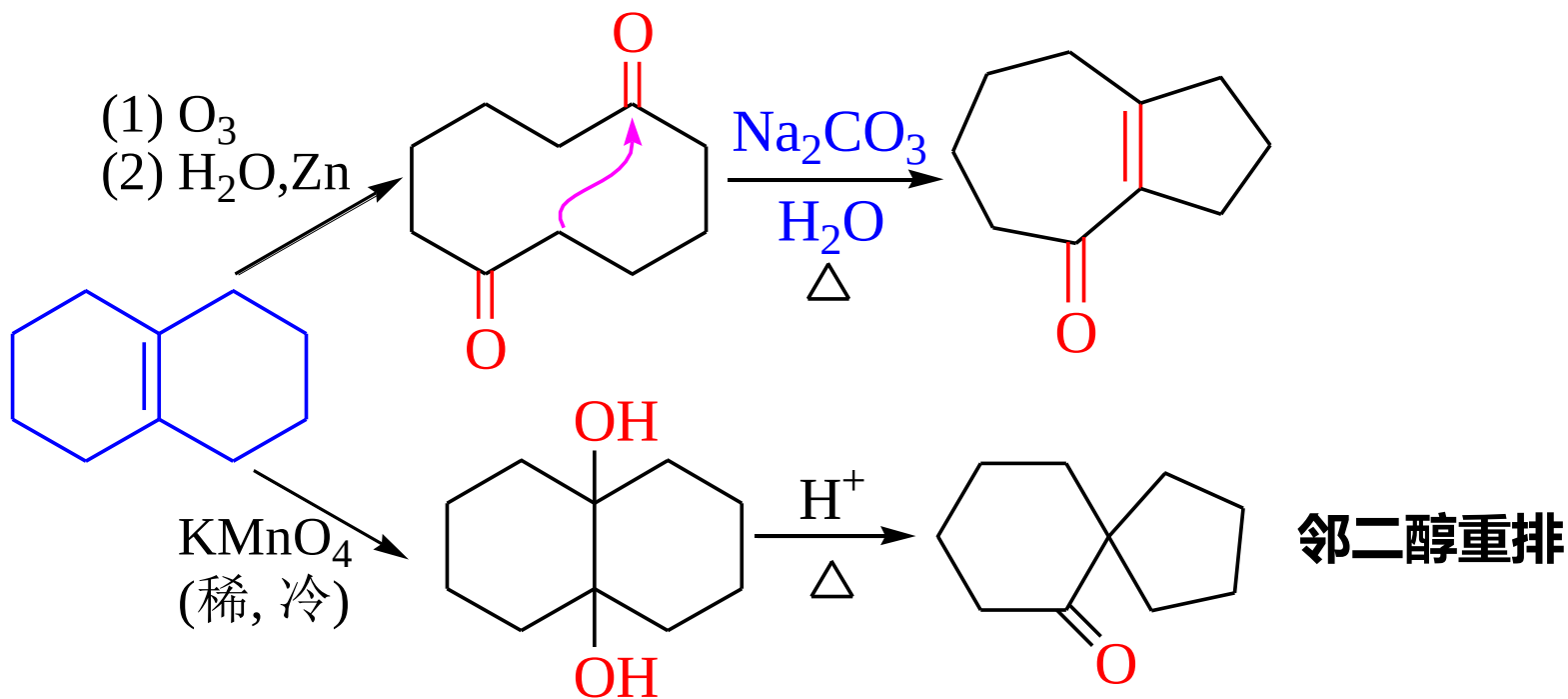
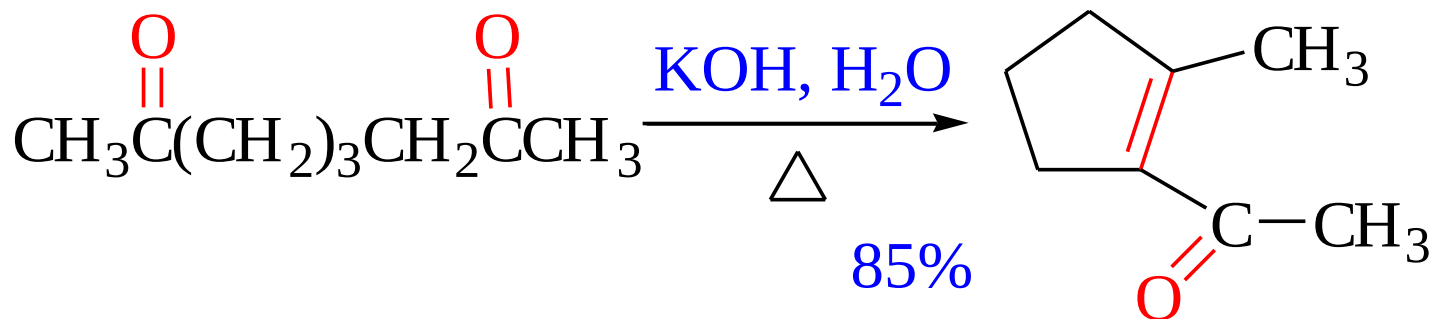
# 反应机理（掌握）



# 分子间缩合

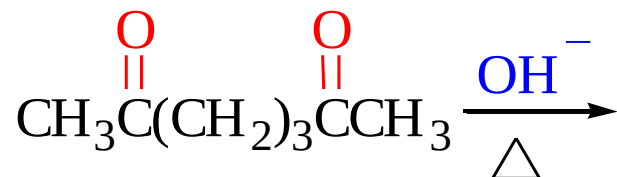
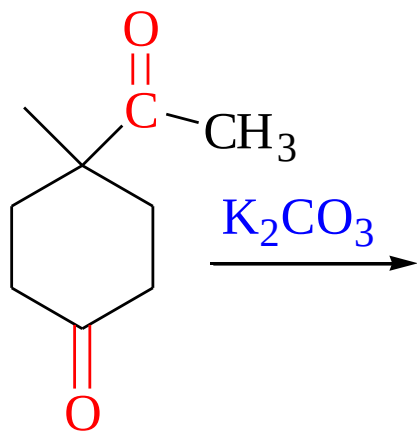


# 分子内缩合



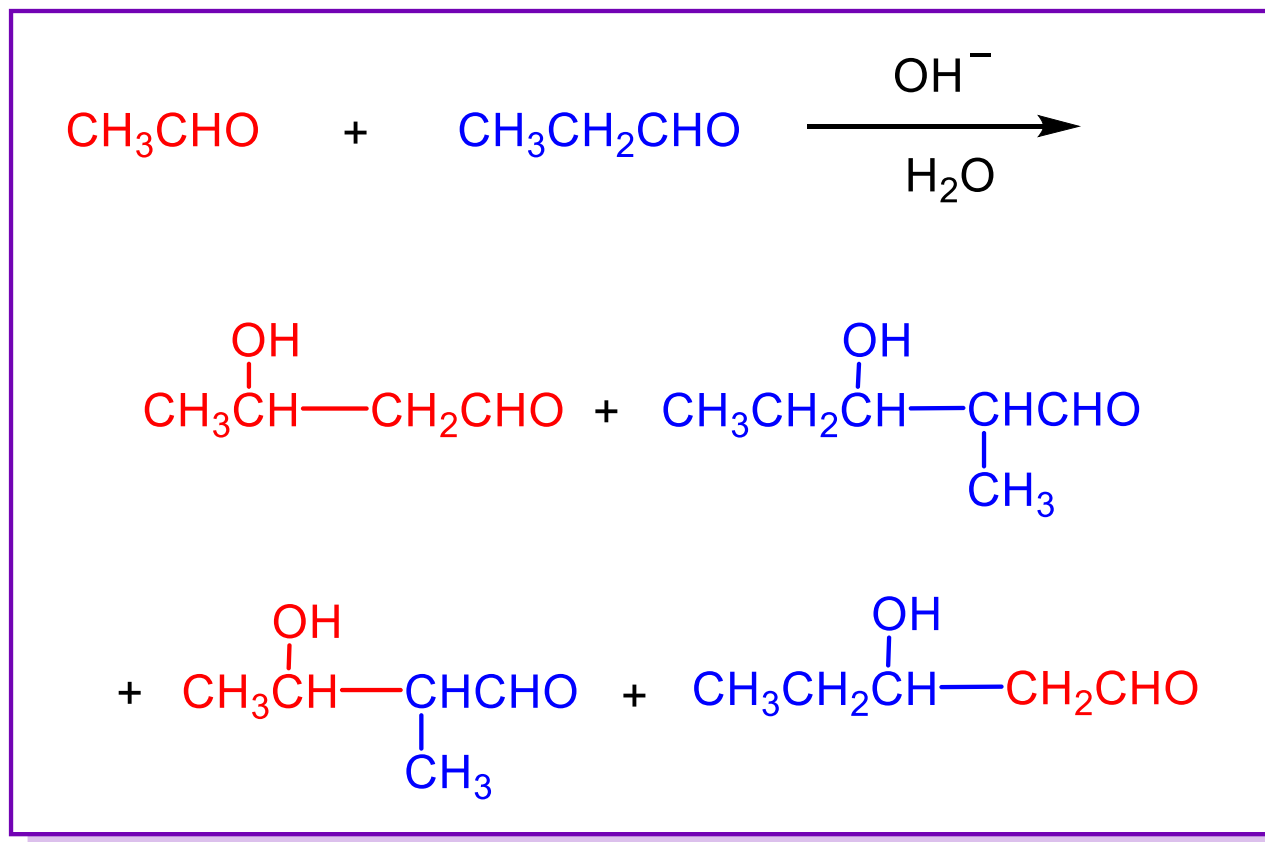
# 测试

请同学们完成下列反应



## 2. Cross aldol additions/condensations

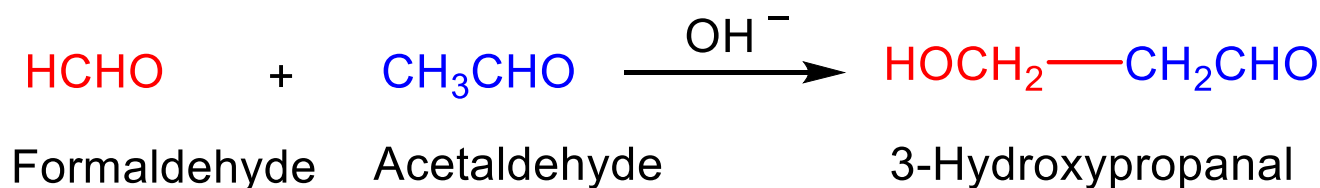
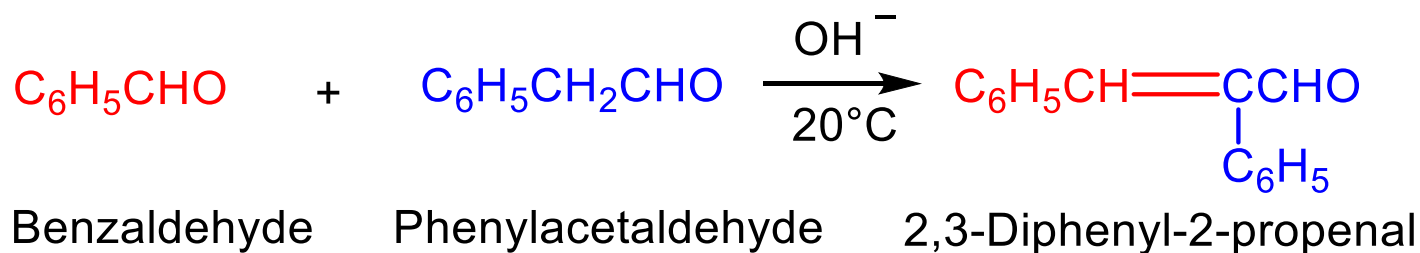
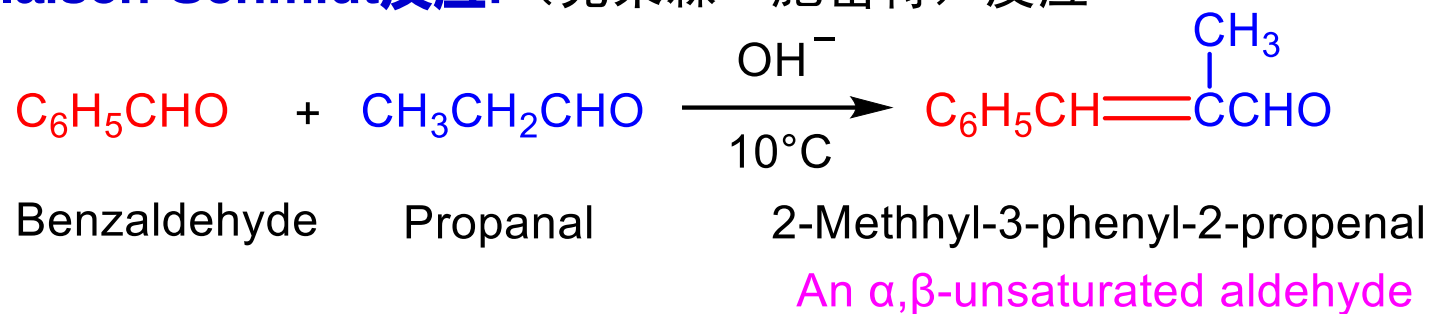
交叉缩合



多种产物生成，包括相应的消除产物 (aldol condensation)

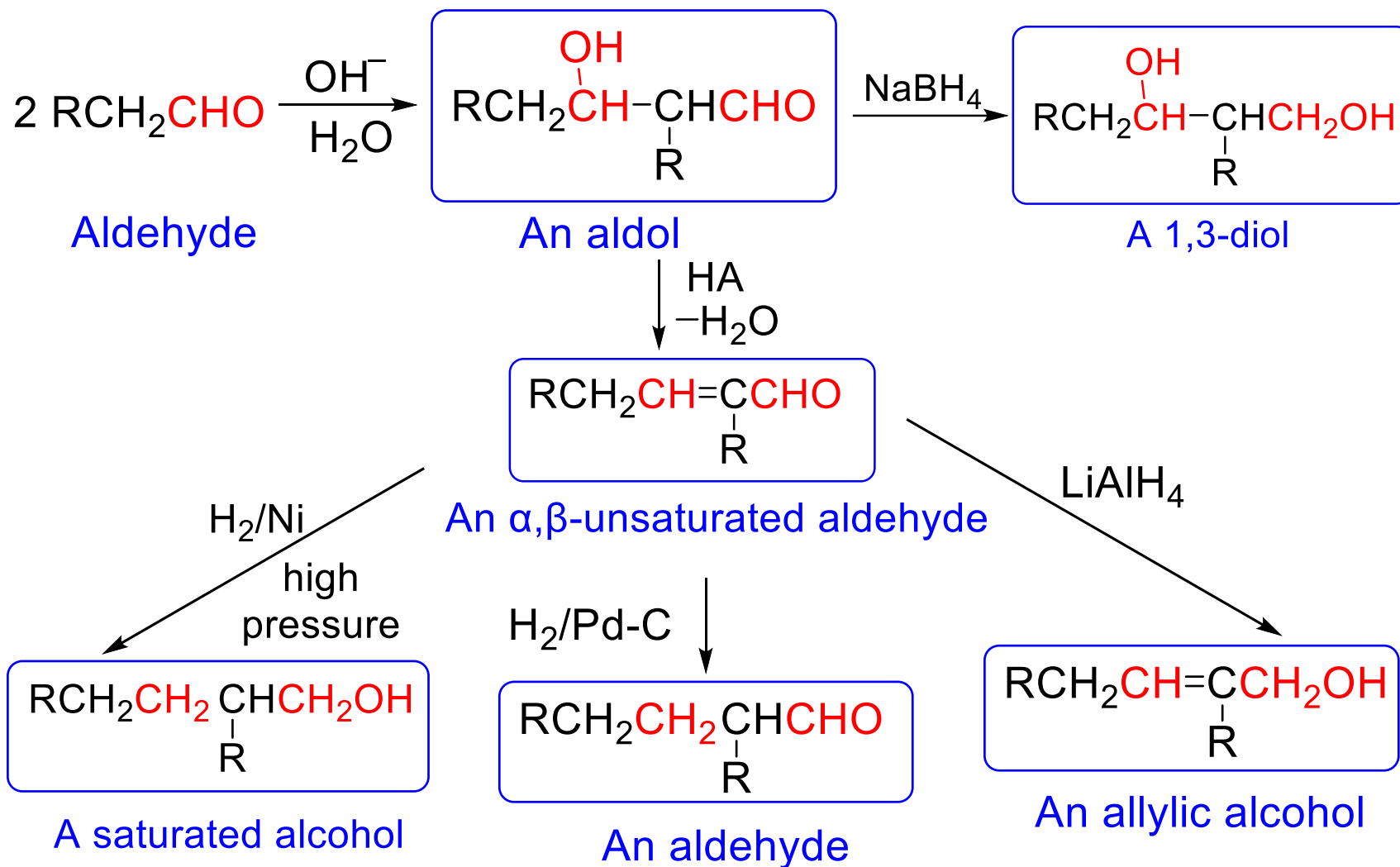
# Practical Cross Aldol Reactions

**Claisen-Schmidt反应:** (克莱森-施密特) 反应



**反应的选择性**

# Synthetic Applications of Adol Products





# 醛、酮的氧化还原反应

## 1. 氧化反应

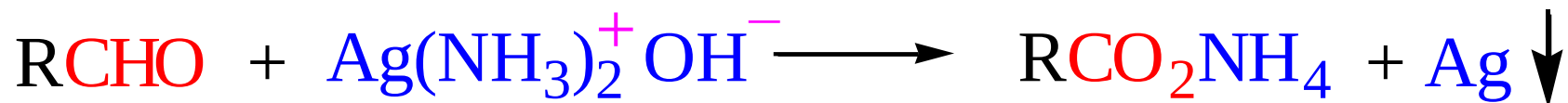
### 醛的氧化



氧化剂:  $\text{KMnO}_4$ ,  $\text{K}_2\text{Cr}_2\text{O}_7$ ,  $\text{H}_2\text{Cr}_2\text{O}_7$ ,  $\text{H}_2\text{CrO}_4$ ,  
 $\text{RCO}_3\text{H}$ ,  $\text{Ag}_2\text{O}$ ,  $\text{H}_2\text{O}_2$ ,  $\text{Br}_2\text{-H}_2\text{O}$ ,

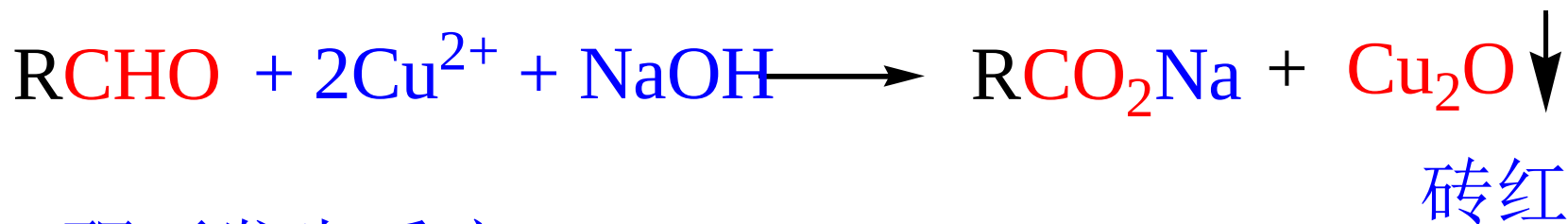
# 可用于鉴别醛、酮的氧化剂

## 1) Tollens 试剂 (银镜反应)



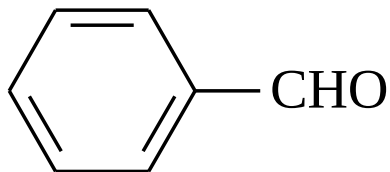
酮不发生反应 (脂肪醛和芳香醛可以发生反应)

## 2) Fehling试剂



酮不发生反应

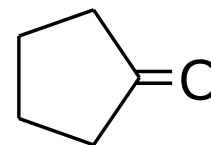
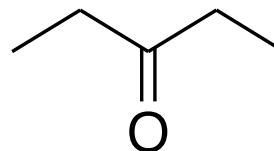
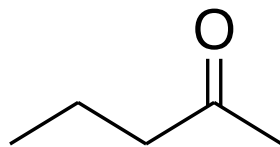
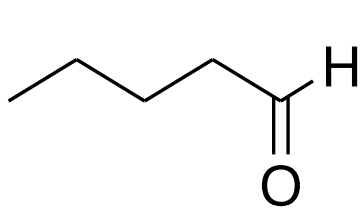
芳香醛不被Fehling试剂氧化



不被Fehling试剂氧化

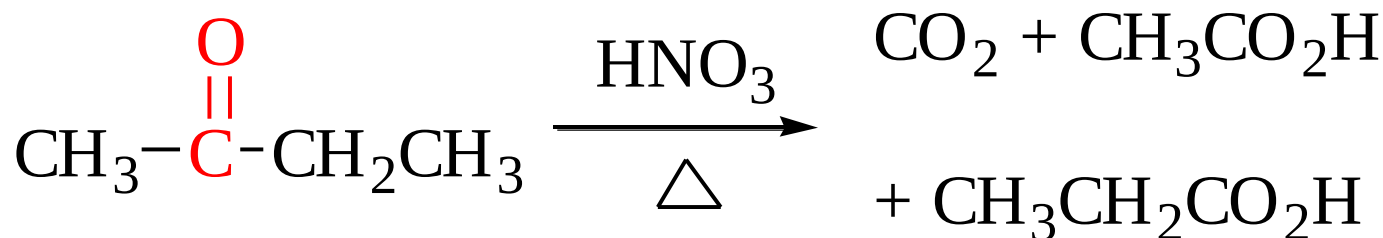
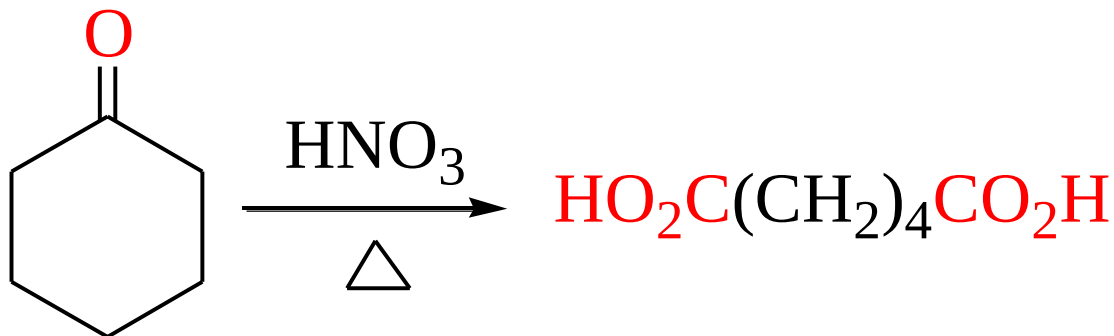
# 测 试

鉴别下列化合物



# 酮的氧化

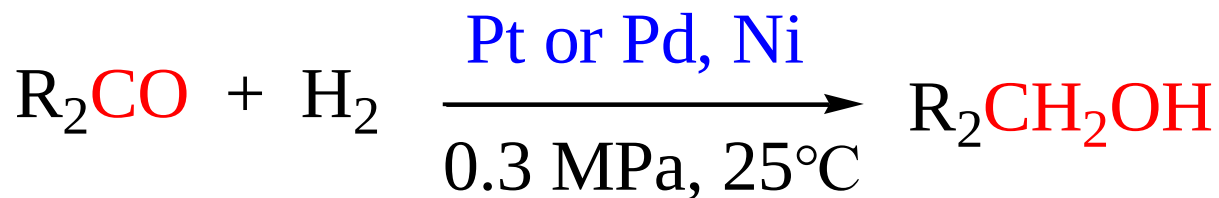
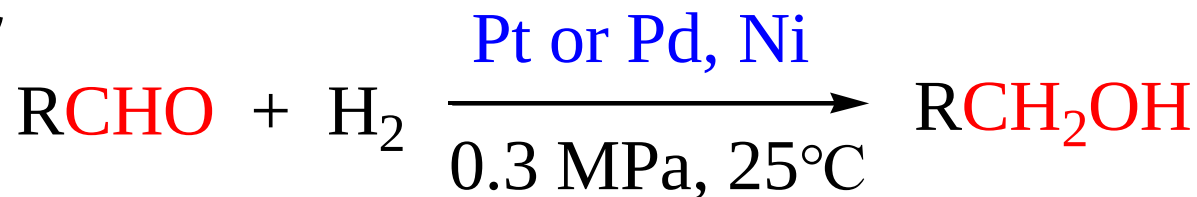
- 酮遇一般氧化剂时，不被氧化
- 酮遇强氧化剂时，发生碳链断裂，氧化成酸



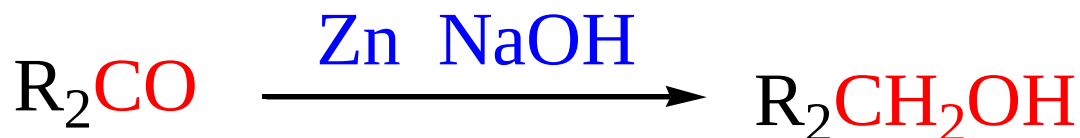
# 醛、酮的还原反应

## 2. 还原为羟基

### 催化氢化

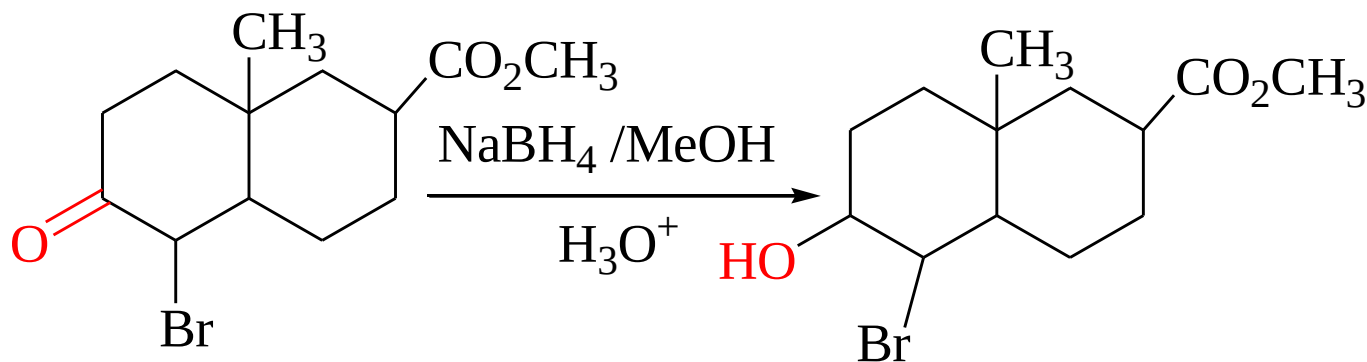
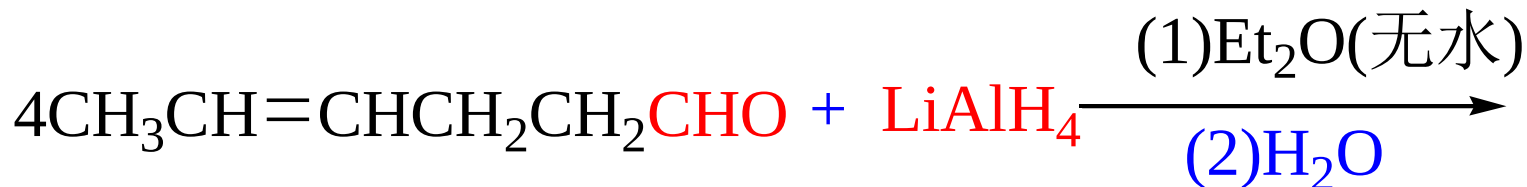


### 金属还原剂



### 3. 用金属氢化物还原

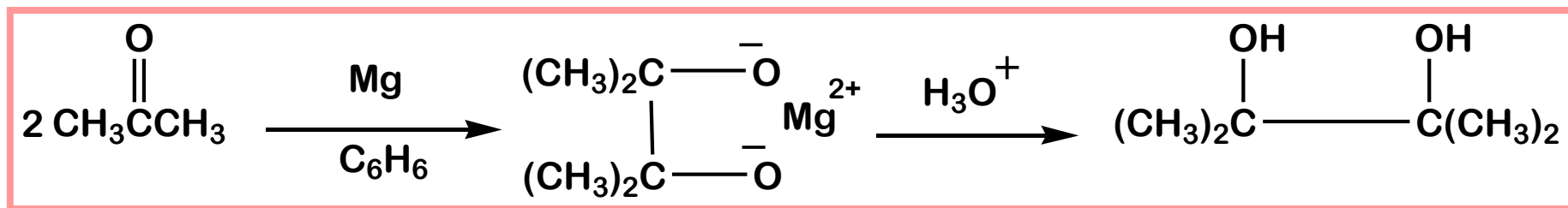
$\text{LiAlH}_4$   $\text{NaBH}_4$



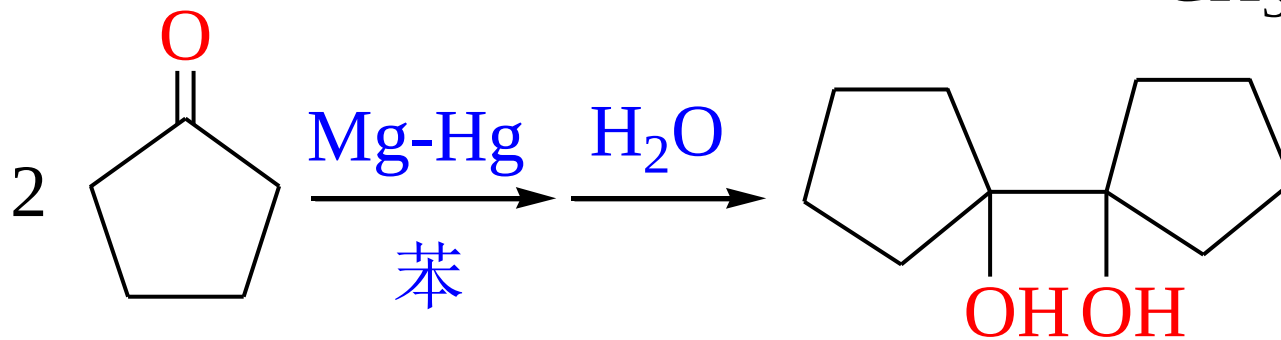
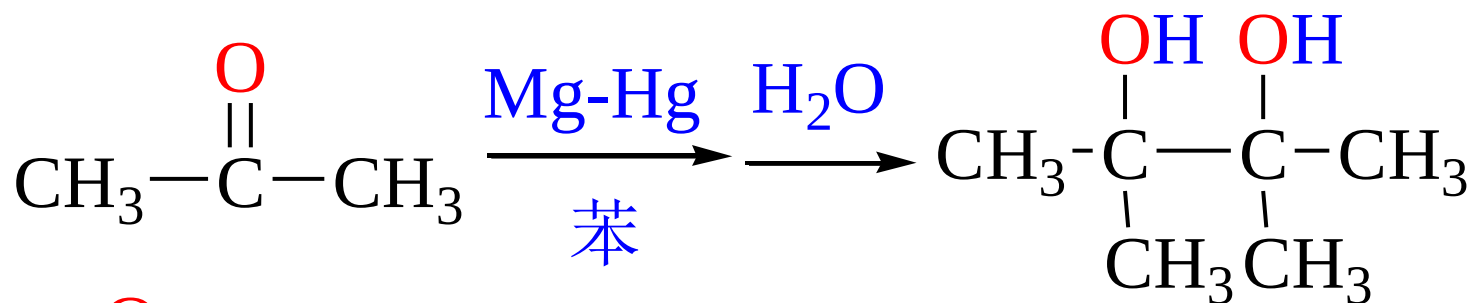
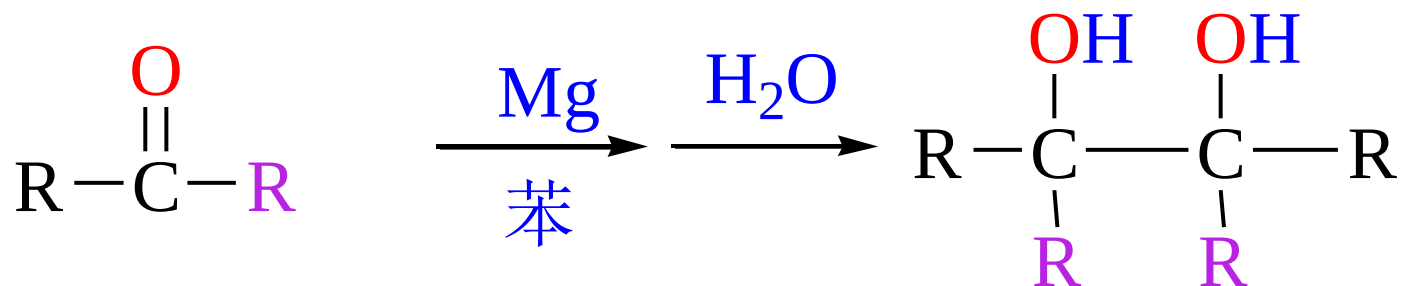
# 醛、酮的氧化还原反应

## (4) 双分子还原

酮与镁、镁汞齐或铝汞齐在非质子溶剂中还原后水解，产物为邻二叔醇



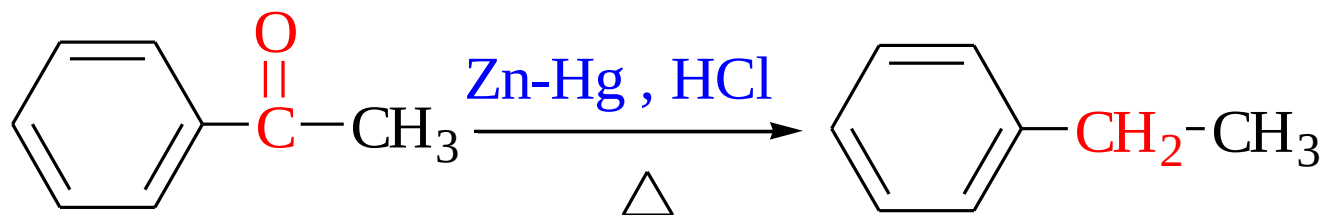
# 双分子还原



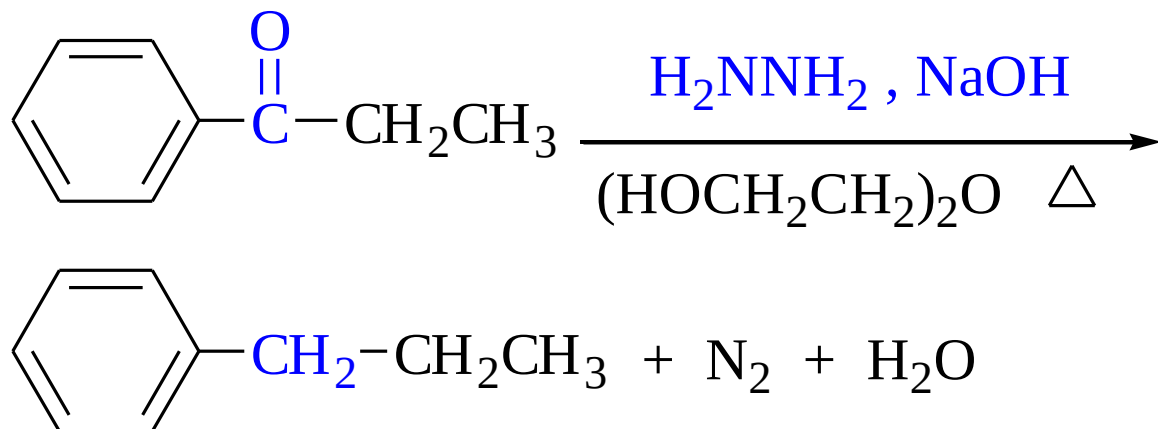


# 还原为亚甲基或甲基

## (1) Clemmensen Reduction



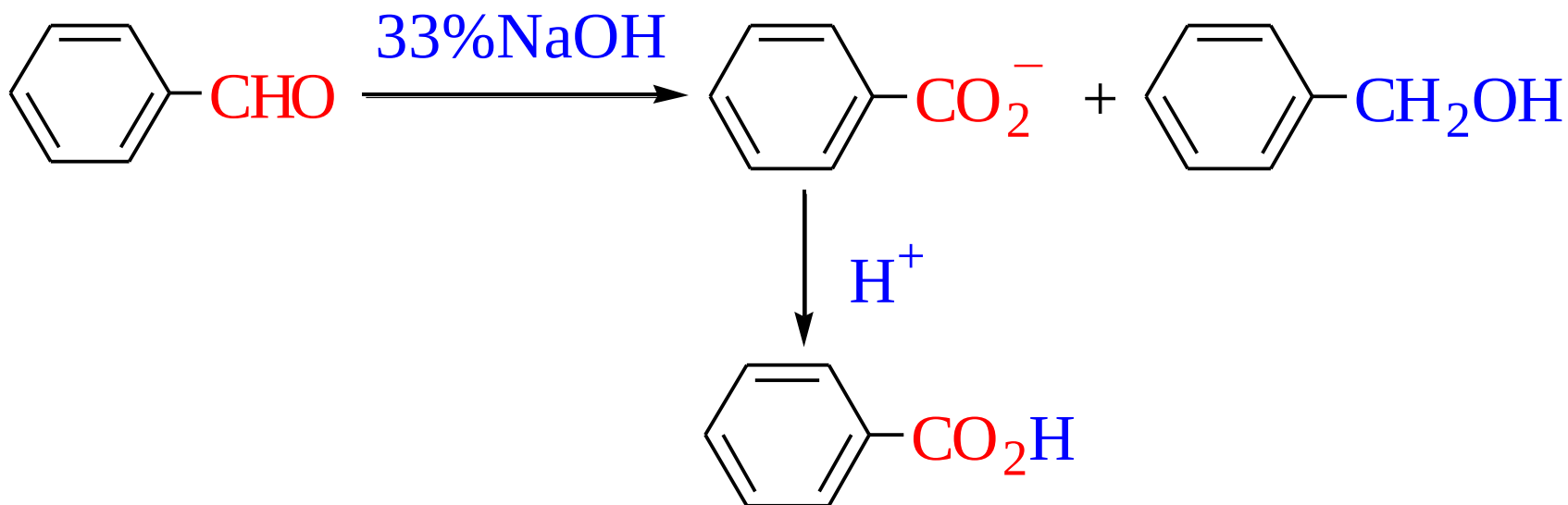
## (2) Wolff-Kishner-黄鸣龙还原



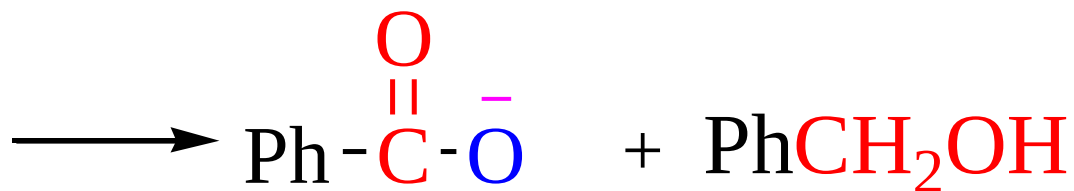
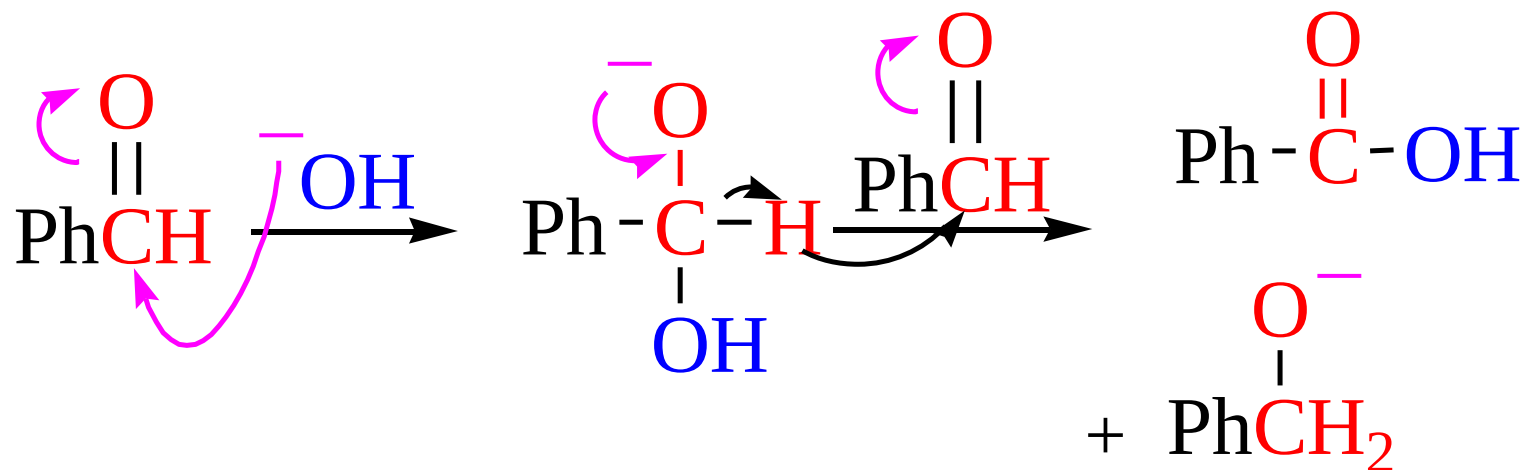
# 岐化反应 Cannizarro 反应

## 岐化反应 Cannizarro reaction

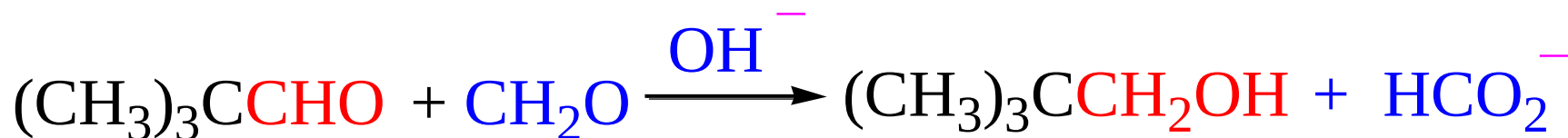
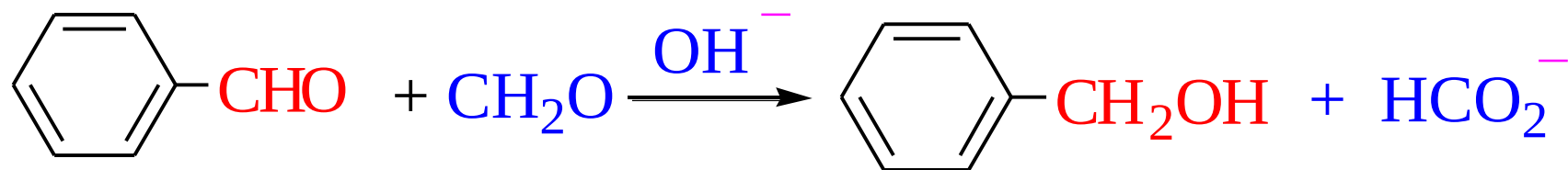
不含 $\alpha$ -H的醛在浓碱作用下发生分子间的氧化—还原反应生成相应的醇和酸



# 反应机理

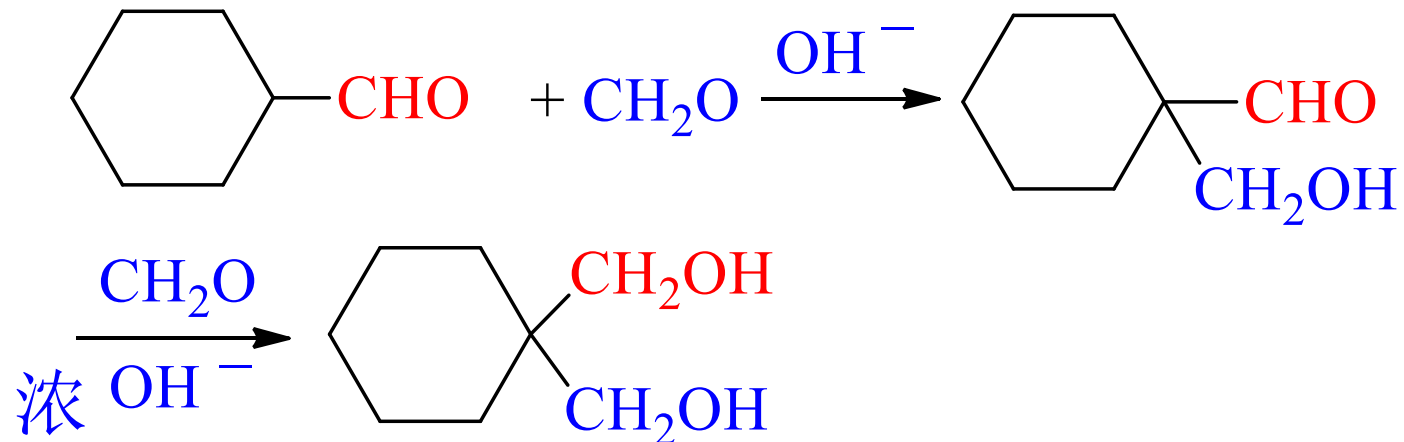
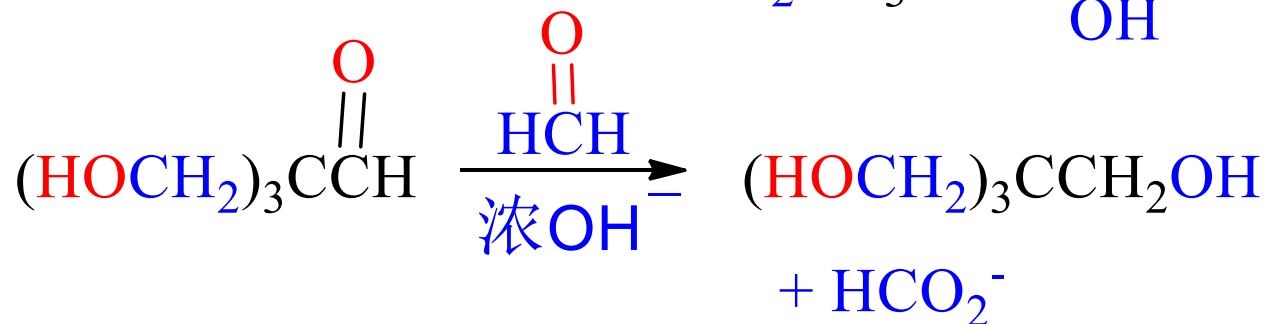
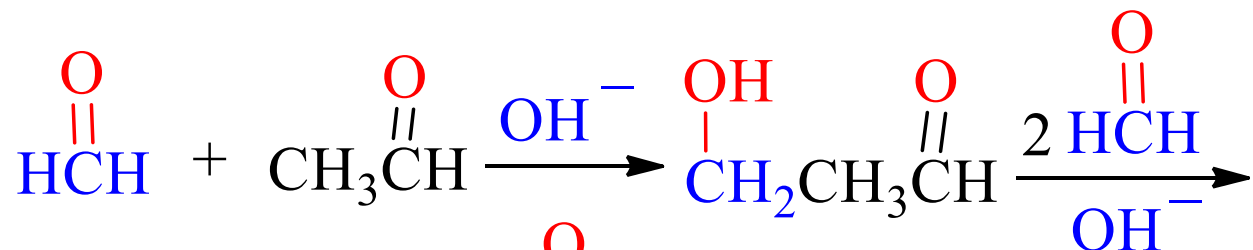


# 交叉的Cannizzaro 反应



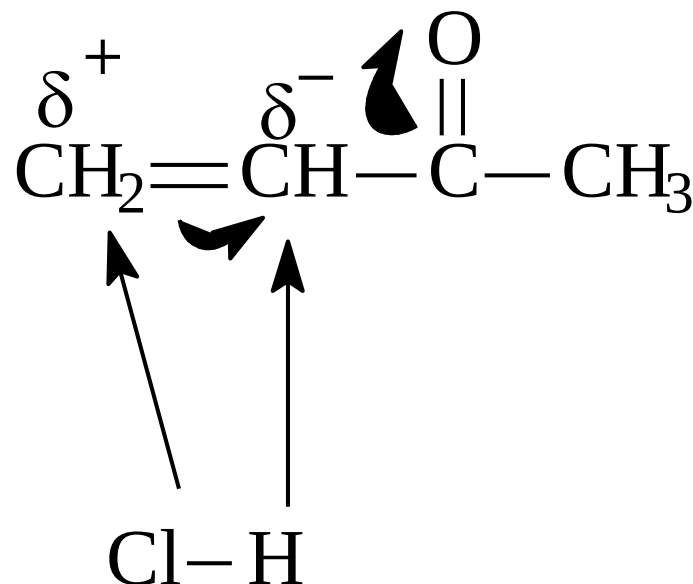
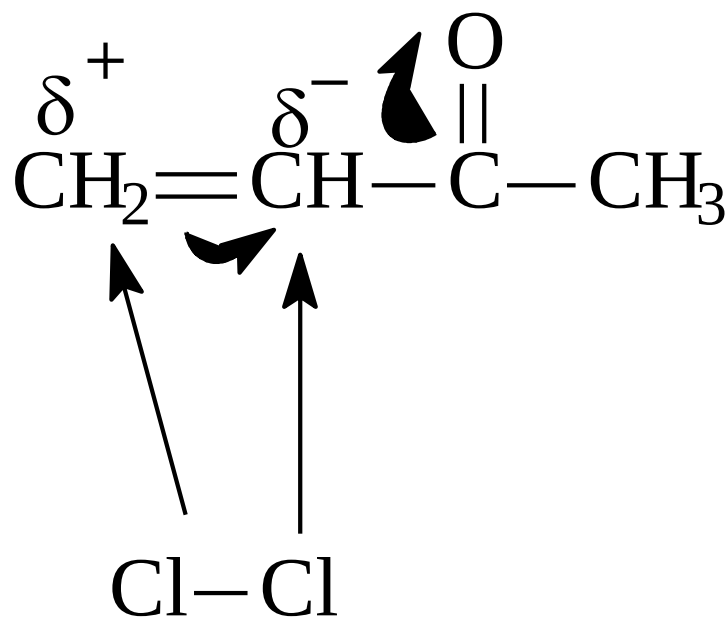
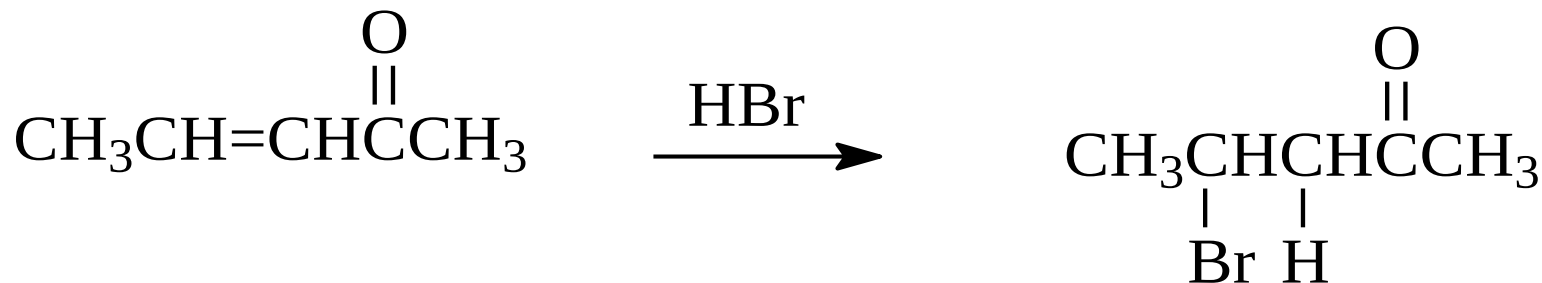
OH<sup>-</sup>首先进攻**正电性较强**的羰基碳

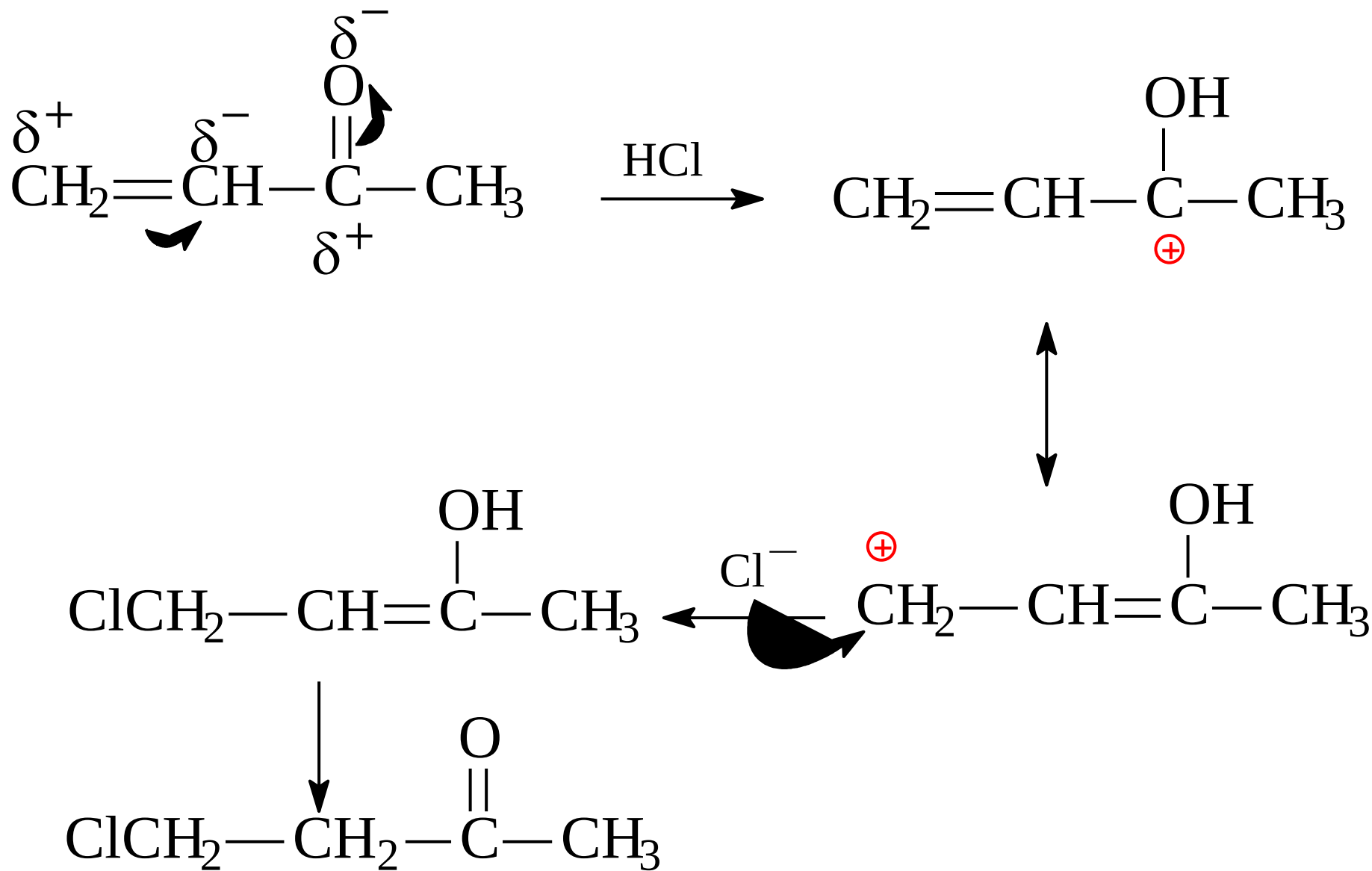
# Example



# $\alpha,\beta$ -不饱和醛酮

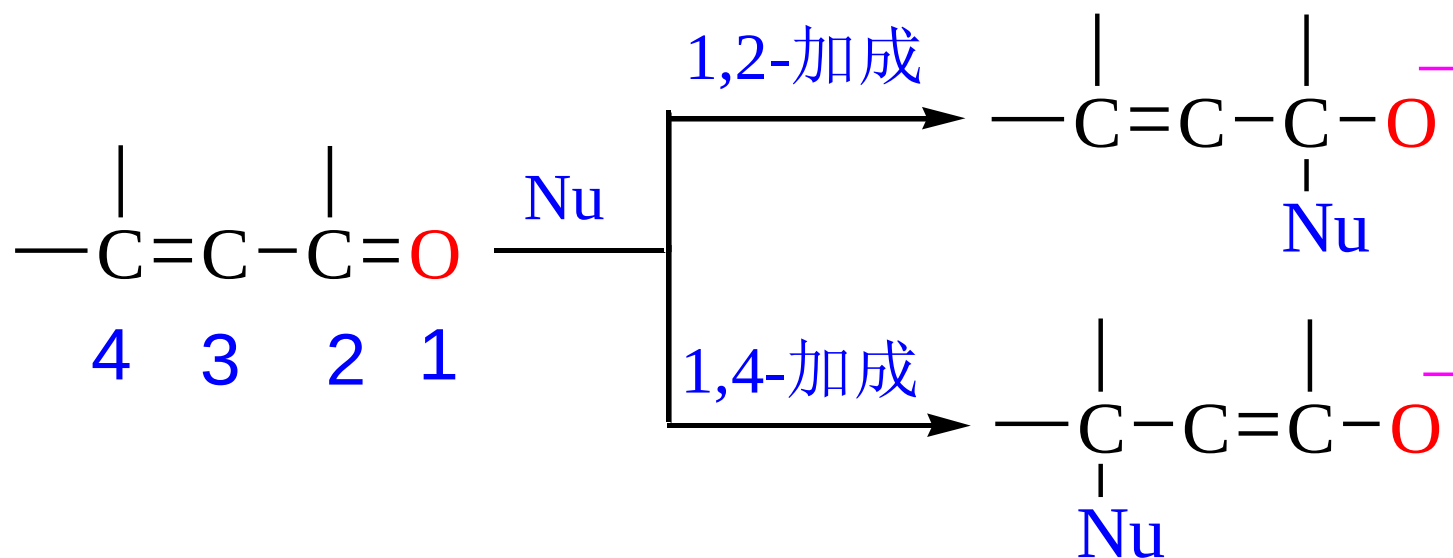
## 1. 亲电加成反应





# $\alpha,\beta$ -不饱和醛酮

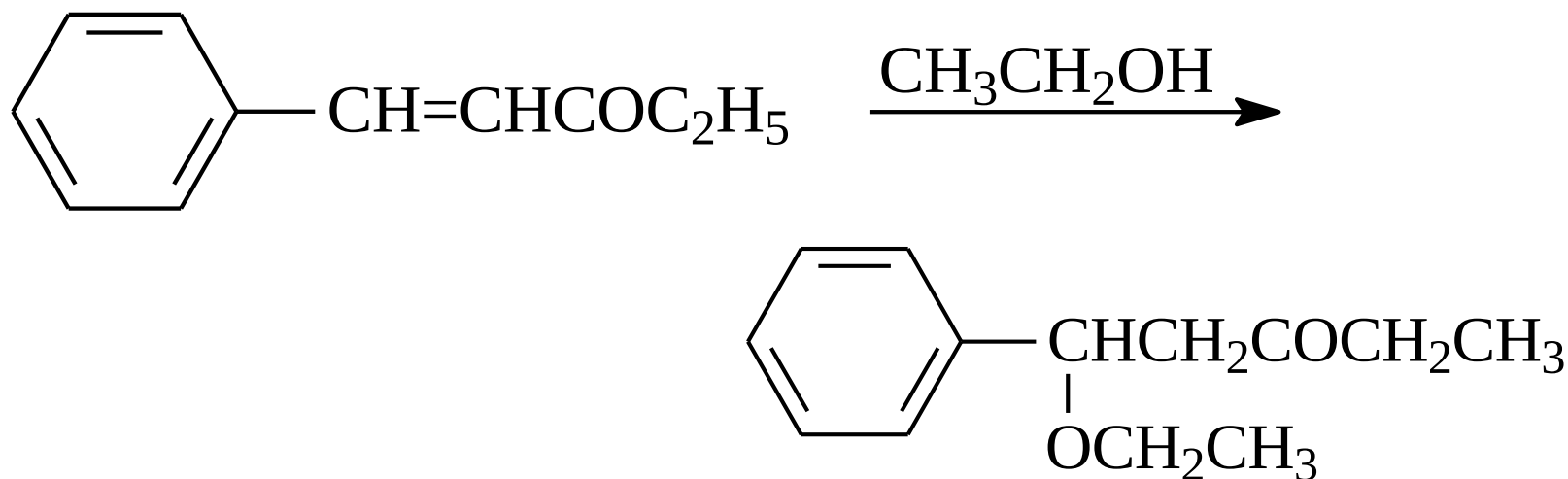
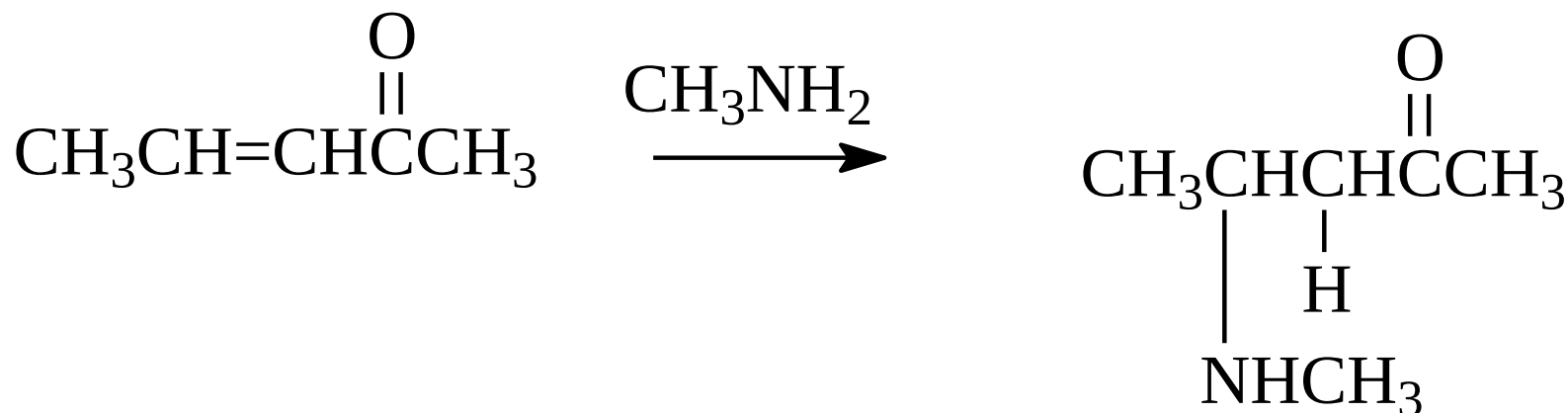
## 2. 亲核加成



一般得到3, 4-加成产物



# Example



# Perkin反应

